

SEEDPRINTS: FINGERPRINTS CAN EVEN TELL WHICH SEED YOUR LARGE LANGUAGE MODEL WAS TRAINED FROM

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ABSTRACT

The fingerprinting of Large Language Models (LLMs) constitutes an indispensable prerequisite for provenance verification and model attribution. Prevailing fingerprinting methodologies are evaluated predominantly in the post-fine-tuning regime, wherein models have already accreted stable signatures derived from training data, optimization dynamics, or hyperparameter configurations. Yet, since the preponderance of a model’s capacity and encoded knowledge is acquired during pretraining rather than through downstream fine-tuning, large-scale pretraining constitutes a more foundational regime for lineage verification. We demonstrate that extant fingerprinting methods grow *unreliable* in precisely this regime, insofar as they are predicated upon post-hoc signatures that materialize only after substantial optimization—a limitation that stands in direct tension with the classical Galtonian conception of a fingerprint as an intrinsic and enduring mark of identity. As a corrective, we advance a stronger and more intrinsic notion of LLM fingerprinting: **SeedPrints**, a method that exploits the biases of random initialization as persistent, seed-dependent identifiers that are inscribed even prior to the commencement of training. We establish that untrained models manifest reproducible prediction biases induced by their initialization seed, and that these attenuated signals remain statistically detectable across the entirety of training, thereby affording high-confidence lineage verification. In contradistinction to prior techniques, which either fail outright during early pretraining or degrade under distributional shift, **SeedPrints** retains its efficacy across all training stages, spanning initialization, large-scale pretraining, and downstream adaptation. Experiments conducted on LLaMA-style and Qwen-style models exhibit seed-level distinguishability and enable birth-to-lifecycle identity verification. Evaluations on large-scale pretraining trajectories and real-world fingerprinting benchmarks further corroborate its robustness under protracted training, domain shifts, and parameter modifications. Taken together, our findings evidence that initialization itself imprints a singular and persistent identity upon LLMs, constituting a genuine “Galtonian” fingerprint. Code will be released upon acceptance.

1 INTRODUCTION

LLM fingerprints have recently been advanced as an instrument for identifying, attributing, and tracing LLMs through the scrutiny of their observable behaviors (Pasquini et al., 2024; Xu et al., 2024; Yoon et al., 2025; Zhang et al., 2024; Zeng et al., 2024). Such methods aspire to furnish model owners with a verifiable evidentiary link between a suspicious model and its putative progenitor, thereby enabling the detection of model theft or unauthorized reuse (Yoon et al., 2025; Zhang, 2025).

A substantial portion of this literature explicitly appropriates the metaphor of biological fingerprints from Francis **Galton**’s *Finger Prints* (1892) (Galton, 1892):

“A fingerprint is the pattern formed by friction-ridge skin on the fingertips; this ridge configuration is individually unique and essentially permanent across an individual’s lifetime.”

SeedPrints vs Baselines: OLMo-2-7B Pre-training Checkpoints

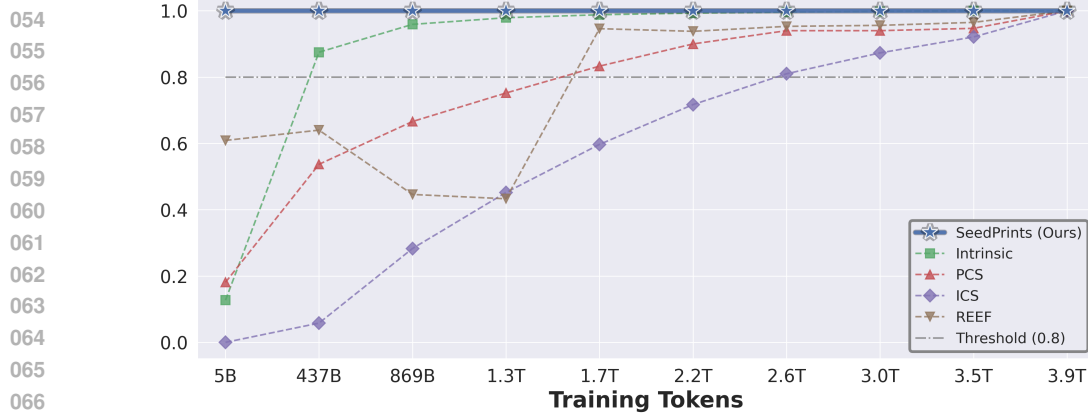


Figure 1: **Existing fingerprinting methods fail to detect model lineage during early pre-training.** We juxtapose five methods on OLMo-2-7B checkpoints spanning 5B to 3.9T training tokens, each evaluated against the terminal checkpoint. The y-axis reports the similarity score, for which higher values betoken a stronger lineage signal, while the dashed line demarcates the 0.8 detection threshold. Whereas all baselines deteriorate and subside beneath the threshold at the earliest checkpoints, **SeedPrints** attains flawless detection ($p \ll 0.001$, plotted as $1 - p$) from the very first checkpoint onward.

The analogy intimates that an effective fingerprint ought to be at once unique and persistent, discernible from the very instant of a model’s “birth” at initialization. Yet the majority of extant so-called fingerprinting approaches fall demonstrably short of this desideratum (Pasquini et al., 2024; Xu et al., 2024; Yoon et al., 2025; Zhang et al., 2024; Zeng et al., 2024; Zhang, 2025; Luan et al., 2025; Tsai et al., 2025; Alhazbi et al., 2025). As illustrated in Figure 1, existing methods prove incapable of reliably detecting lineage during early pretraining. This failure stems from the fact that such methods are operationalized only once models have fully converged—for instance, by extracting patterns from parameters or from generated text. In consequence, the separability they achieve reflects not a birthmark intrinsic to the model itself, but rather an imprint of surface-level training constituents such as data signatures, optimization dynamics, or hyperparameters. To take a concrete illustration, in Section 5.1 existing baselines frequently fail to identify lineage correctly whenever the training data distribution shifts appreciably. Such methods, therefore, operate more as post hoc identifiers than as **Galtonian fingerprints**—those innate, immutable marks that accompany a model from its very inception.

In this work, we articulate a more stringent notion of LLM fingerprinting, conceived as an intrinsic property that is present already at *initialization* and remains detectable at *any* juncture of the ensuing training. Our principal contributions are as follows:

- **Identifying a fundamental limitation of existing methods.** We demonstrate that prior fingerprinting approaches are unable to reliably detect lineage during early pretraining and are readily misled whenever the training data distribution undergoes substantial shifts. (Section 5.1)
- **Discovering and leveraging initialization-driven fingerprints.** We uncover that untrained models already harbor seed-dependent biases within their internal representations, biases that constitute a weak yet persistent identity signal present from the moment of initialization. (Section 3) Building upon this observation, we introduce **SeedPrints**, a method that isolates these initialization-born fingerprints and exhibits markedly greater robustness to confounding factors such as data distribution, training duration, and optimization dynamics. (Section 4)
- **Extensive empirical validation.** We establish that our method discriminates between models differing solely in their initialization seed—even under identical training pipelines and data orderings—and that it persists under continual training across heterogeneous datasets. (Section 5.1)
- **Strong performance in realistic settings.** Our method reliably detects lineage throughout large-scale pretraining trajectories (e.g., OLMo-2-7B Stage 1), without succumbing to early-stage failure. It further performs strongly in large-scale fine-tuning regimes (e.g., up to 700B tokens) and in practical deployment scenarios instantiated by the *LeaFBench* benchmark (Shao et al., 2025), which encompasses 65 models spanning 7 mainstream model families and 6 transformations (namely instruction tuning, fine-tuning, PEFT, quantization, model merging, and distillation). Across these

settings, our method attains parity with the strongest baseline (near-perfect) while substantially surpassing all remaining competitors (Sections 5.2 and 5.3).

2 RELATED WORK

The protection of LLMs broadly bifurcates into two families: (i) *watermarking / active fingerprinting* methods, which *insert* an identifiable signature into a model or its outputs (Xu et al., 2024; Nasery et al.; Tsai et al., 2025; Sander et al., 2024), and (ii) *passive fingerprinting* methods, which *extract* a signature from a model’s pre-existing behaviors without effecting any modification (Yoon et al., 2025; Zhang et al., 2024; Alhazbi et al., 2025; Pasquini et al., 2024; Suzuki et al., 2025; Zhang, 2025).

Watermarking and Active Fingerprinting. Active approaches deliberately implant verifiable identifiers in anticipation of subsequent ownership adjudication. Classic techniques encompass backdoor attacks (Adi et al., 2018; Li et al., 2019; Zhang et al., 2018) as well as digital signatures and hash functions (Guo & Potkonjak, 2018; Zhu et al., 2020). In the case of language models, two prevalent instantiations may be distinguished: *text watermarks*, which bias generation or interpolate predefined patterns so as to encode concealed information (Kirchenbauer et al., 2023; Xu et al., 2024; Nasery et al.); and *model weight watermarks*, which embed identifiers directly into parameters or bind them to secret triggers through fine-tuning (Luan et al., 2025; Li et al., 2023). Although backdoor-style fingerprints are comparatively straightforward and may endure beyond moderate fine-tuning (e.g., Dasgupta et al. 2024), these invasive schemes presuppose control over the training process, a requirement that renders them ill-suited to the retroactive marking of third-party models.

Passive LLM Fingerprinting. Passive fingerprinting, by contrast, identifies models through an analysis of their intrinsic properties, entirely without modification. Passive techniques differ according to the degree of model access they presuppose. Under **white-box access**, signatures are extracted from the model weights themselves, capitalizing on intrinsic properties such as the distribution of attention matrices (Yoon et al., 2025), the kernel alignment of internal representations (Zhang et al., 2024), or the stable directionality of parameter vectors. In the **black-box setting**, fingerprinting instead devolves upon the analysis of input-output behavior; such methods enlist crafted queries (Pasquini et al., 2024), unique prompt-response pairs (Tsai et al., 2025), stylometry (Alhazbi et al., 2025), or iterative prompting–response games (Iourovitski et al., 2024) to identify a model, albeit at the cost of diminished robustness to fine-tuning. Crucially, however, all of these methods define their signatures *post hoc*: they discern emergent properties of a training process already brought to completion, rather than the innate, “Galtonian” fingerprints—present from random initialization—that our work sets out to discover. Consequently, *they are not designed for lineage verification during large-scale pretraining*, precisely the stage at which such post-training signals have not yet crystallized.

3 BIASES ORIGINATING FROM INITIALIZATION PERSIST AFTER TRAINING

In this section, we expound our central observation: language models exhibit pronounced prediction biases that originate in the random initialization seed, and such biases persist detectably even in the aftermath of training.

Initialization induces seed-specific bias profiles. We evaluate a LLaMA-2–style model initialized with seed 123 on 10,000 random input sequences of length 1,024, wherein tokens are drawn uniformly from the vocabulary. In Figure 2 (Left), we plot the frequency with which each output token (upper, derived from the final logits) and each hidden dimension (lower, derived from the last-layer representations) attains the minimum value across the 10,000 random trials. The conspicuously non-uniform profile evident in both plots betrays the presence of extreme output-preference bias patterns. Replicating the experiment across a range of initialization seeds reveals that, while the aggregate magnitude of the bias remains stable, the specific argmin dimensions are contingent upon the seed (Figure 2, Lower Right). These observations accord with recent findings according to which the biases of randomly initialized models arise from inter-sequence representation contraction, itself driven by asymmetric nonlinear activations within MLP blocks and further amplified by self-attention (Li et al., 2026). On this account, distinct initialization seeds engender distinct contraction directions, which in turn yield distinct argmin patterns across the output dimensions.

Training preserves initialization-born bias Although training substantially reconfigures output magnitudes, the relative preference over output dimensions instilled at initialization is by no means wholly effaced. We train the previously initialized model (seed 123) for a single epoch on OpenWeb-

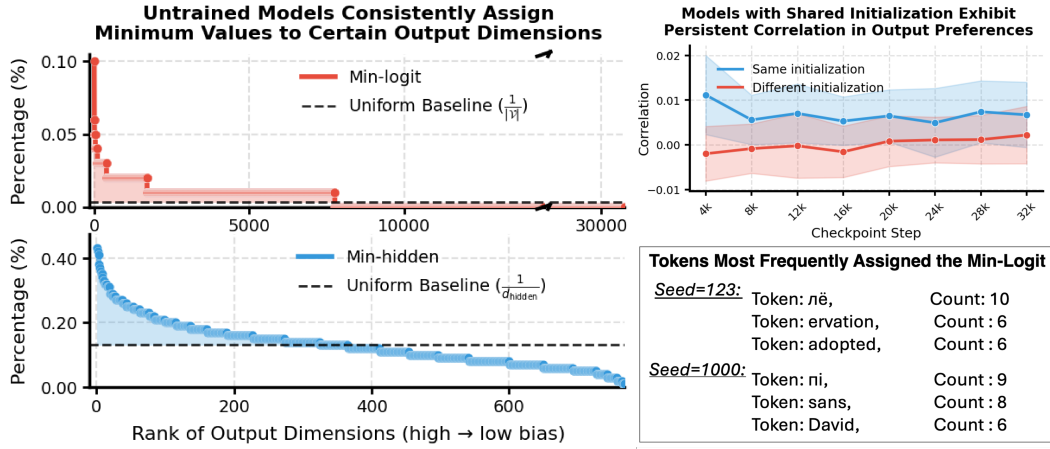


Figure 2: **Initialization-born output bias persists through training.** *Left:* Confronted with wholly random inputs, the outputs of a randomly initialized LLaMA-2-style model are far from uniform and instead evince a pronounced bias, whereby certain dimensions are systematically disfavored by the model (i.e., they frequently receive the minimum value across random inputs). Such extreme bias surfaces both in the logits (top, red) and in the final hidden representations (bottom, blue). The dashed line denotes the frequency to be expected under a uniform distribution, while the arrows in the top panel signal a broken x-axis that elides low-frequency tail ranks. *Upper Right:* Over the course of training, models remain weakly correlated in their output bias, across inputs, with the base model from which they descend. This correlation distribution is consistently displaced upward relative to correlations computed against independently initialized base models, an observation indicating that initialization-born bias leaves a measurable signal enduring through training. *Lower Right:* The specific dimensions that manifest strong bias are contingent upon the initialization seed; here we illustrate this phenomenon using the tokens that most frequently receive the minimum logit.

Text (Gokaslan et al., 2019) and subsequently evaluate its intermediate checkpoints. We concentrate on the $m = 50$ output dimensions that the initialized model most frequently assigns minimum values. For each checkpoint, we compute the correlation between its responses and those of the initialized model across random inputs, restricted to precisely these dimensions. Intuitively, this quantifies the extent to which the checkpoint and the initialized model rank, or disfavor, these dimensions congruently across inputs. As a point of reference, we compute the identical correlation against other, independently initialized models (seed 1000). As displayed in Figure 2 (Upper Right), notwithstanding the smallness of the absolute correlation values, the correlations for models sharing an initialization are consistently displaced upward relative to the independent baseline, and they stabilize rather than decaying toward zero. Taken as a whole, this indicates that initialization bequeaths a weak yet statistically detectable bias signal that endures throughout training.

4 ALGORITHM

Section 3 establishes that models descending from a common initialization lineage retain a weak yet consistent agreement over a subset of output dimensions (e.g., those most strongly disfavored by the untrained model under random inputs). Our objective is accordingly to uncover, from any two given models—and without recourse to the true initialized model—a set of dimensions that encode such bias signals, and thereafter to adjudicate whether their agreement is statistically significant. The overall procedure is summarized in Algorithm 1, and we delineate each of its components below.

Extract identity dimensions between any two models. Let $X = \{x_i\}_{i=1}^n$, wherein each $x_i \in \mathbb{R}^{\ell \times d}$ denotes a random sequence of length ℓ . Each random input x_i admits instantiation either by uniformly sampling ℓ token IDs from the vocabulary or, alternatively, by directly sampling ℓ independent vectors from a d -dimensional isotropic Gaussian distribution. For any model g , we define its mean response vector as $\bar{g} := \frac{1}{n} \sum_{i=1}^n g(x_i) \in \mathbb{R}^{d_{\text{out}}}$, where d_{out} denotes the output dimensionality. Here, $g(x_i)$ may designate either the model’s logits or its final-layer representations. We then identify its disfavored dimensions as the set of bottom- m coordinates of $\bar{g}$:

$$\mathcal{M}_g := \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \bar{g}_j.$$

We employ the mean response vector to extract bias signals because it furnishes a more stable and noise-robust alternative to per-sample argmin frequency (i.e., the tally of how often each dimension attains the minimum value, as illustrated in Figure 2), while additionally performing well in practice. A formal comparison is deferred to Appendix C.1.

Given two models f and f' , we define their *identity dimensions* as the intersection $\mathcal{S} := \mathcal{M}_f \cap \mathcal{M}_{f'}$. Intuitively, should two models share a common initialization lineage, they ought to identify independently a similar set of disfavored dimensions, thereby producing a non-trivial overlap in their bottom- m sets. This intersection-based criterion accordingly operates as a form of mutual verification: it suppresses spurious correlations attributable to architectural affinity or shared training dynamics, and it isolates those bias signals more plausibly ascribable to a shared initialization lineage. Whenever one of the models lies closer to initialization, its disfavored dimensions may moreover be adopted directly as an anchor.

Correlation statistics. Restricting both models to the identity dimensions $\mathcal{S}$, we collate their responses across random inputs.¹ For each $j \in \mathcal{S}$, we compute a Kendall–Tau rank correlation:

$$\tau_j = \text{KendallTau}(\{f(x_i)_j\}_{i=1}^n, \{f'(x_i)_j\}_{i=1}^n).$$

This procedure yields a set of correlation statistics $T = \{\tau_j\}_{j \in \mathcal{S}}$, which captures the distribution of agreement between the two models across the identity dimensions. As observed in Section 3, models sharing a lineage exhibit distributionally higher agreement on these dimensions than do unrelated models.

Hypothesis testing via null distribution. Because the correlation distributions for same-lineage and different-lineage models may overlap, a simple threshold imposed on individual statistics does not suffice. Instead, we test whether the observed correlations are significantly greater than would be anticipated under the null hypothesis positing no lineage relation. Ideally, this null ought to be estimated from the correlation distribution between independent models, as in Figure 2. Obtaining such a baseline empirically, however, necessitates comparisons across a great many independently initialized models—an undertaking that is computationally onerous. As demonstrated in Appendix B, the null correlation distribution between independently initialized models is well approximated by a Gaussian. This observation naturally motivates a less costly simulation-based alternative, namely the construction of an empirical null distribution using Gaussian surrogates passed through the identical pipeline, as summarized in Algorithm 1.

While this simulation-based null already affords a strong empirical approximation, it nonetheless introduces superfluous randomness and computational overhead: its precision is circumscribed by the number of simulation trials, its results are liable to vary across runs, and the comparison of two empirical distributions typically demands additional assumptions from the adopted test, such as the one-sided t-test or the Mann–Whitney U test. Since the null distribution of Kendall–Tau (under weak independence) is available in closed form, we further supplant this simulation-based procedure with an analytical test, whilst providing both empirical and analytical implementations within our code repository. Concretely, under the null hypothesis, each Kendall–Tau correlation is approximately zero-mean with known variance σ^2 . By the central limit theorem, as the number of identity dimensions $|\mathcal{S}|$ grows, the distribution of $\bar{\tau} = \frac{1}{|\mathcal{S}|} \sum_{j \in \mathcal{S}} \tau_j$ is well approximated by $\bar{\tau} \sim \mathcal{N}\left(0, \frac{\sigma^2}{|\mathcal{S}|}\right)$. Although softmax normalization introduces weak dependencies across dimensions, these effects are attenuated by the use of a comparatively high temperature ($T = 10$), which renders the independence assumption a reasonable approximation in practice. We validate empirically that this Gaussian approximation closely coincides with the distribution obtained from the full pipeline (Appendix B). We therefore compute the one-sided z-score $z = \frac{\bar{\tau}}{\sigma/\sqrt{|\mathcal{S}|}}$ together with the corresponding p-value $p = 1 - \Phi(z)$, in which Φ denotes the standard normal cumulative distribution function. We reject the null hypothesis and declare shared lineage whenever $p < 0.01$.

Practical instantiation. Model outputs may be construed as either logits or hidden representations. Although both choices are congruent with our formulation, the use of logits necessitates markedly more random inputs to estimate bias patterns reliably as the vocabulary size grows, thereby incurring

¹Prior to computing the correlations, we apply a row-wise softmax normalization to each sample’s output vector. This expedient removes scale disparities across models and dimensions while conserving the relative ordering that conveys identity information for rank-based statistics such as Kendall–Tau. A comparatively high temperature is employed so as to forestall degenerate one-hot behavior.

Algorithm 1 Distribution Correlation Test on Identity Dimensions

Require: base model f , suspicious model f' ; random inputs $X = \{x_i\}_{i=1}^n$; fingerprint size m ; significance level α

► **Step 1: Localize biased dimensions**

- 1: Compute average outputs $\bar{f}, \bar{f}'$ over X
- 2: $\mathcal{M}(f), \mathcal{M}(f') \leftarrow$ bottom- m dimensions of $\bar{f}, \bar{f}'$
- 3: $\mathcal{S} \leftarrow \mathcal{M}(f) \cap \mathcal{M}(f')$ (identity dimensions)

► **Step 2: Form correlation distribution**

- 4: Normalize $f(x_i)_{\mathcal{S}}, f'(x_i)_{\mathcal{S}}$ for each $x_i \in X$ across identity dimension
- 5: **for** each $s_j \in \mathcal{S}$ **do**
- 6: $\tau_j \leftarrow \text{KendallTau}(\{f(x_i)_{s_j}\}_{i=1}^n, \{f'(x_i)_{s_j}\}_{i=1}^n)$ (correlation in in input-wise preferences)
- 7: **end for**
- 8: $\mathcal{T} \leftarrow \{\tau_j\}_{j=1}^{|\mathcal{T}|}$

► **Step 3: Hypothesis test against null**

- 9: Construct $\mathcal{T}_{\text{null}}$ by applying the same pipeline to two Gaussian matrices $Y^{(1)}, Y^{(2)} \sim \mathcal{N}(0, I)^{N \times d_{\text{out}}}$
- 10: Test $H_0 : \mathcal{T} = \mathcal{T}_{\text{null}}$ vs. $H_1 : \mathcal{T} > \mathcal{T}_{\text{null}}$
- 11: **Return** SameLineage $\leftarrow \mathbf{1}(p\text{-value} < \alpha)$

substantial computational overhead (see Appendix C.2 for the theoretical analysis). In the interest of efficiency and stability, we therefore adopt final-layer hidden states as the default model output. The resulting p-value furnishes a direct statistical test for lineage, obviating any reliance upon heuristic similarity thresholds.

5 EXPERIMENTS

This section comprises two parts. In Section 5.1, we demonstrate that our method operates as a genuine fingerprint: (i) it enables *birth verification at the seed level*, and (ii) it remains *verifiable across the entire training lifecycle*. Existing baselines, by contrast, are unable to support verification at the early pretraining stages and rely predominantly on data-dependent signals that disintegrate under pronounced distribution shifts. We report results obtained with the analytical null under *hidden state* outputs, and we relegate additional results—based on the empirical null under both *logit* and *hidden state* outputs, and evaluated by means of the one-sided *t*-test and the Mann–Whitney *U* test—to Appendix D.2. In Section 5.3, we evaluate all methods under practical infringement scenarios using *LeaFBench* (Shao et al., 2025). Across 65 distinct model instances spanning 6 representative post-development techniques, our method attains parity with the strongest baselines in post-training settings while remaining robust to a diversity of deployment transformations.

Baselines We principally consider four passive fingerprinting baselines, whether weight- or representation-based. *Intrinsic fingerprint* (Yoon et al., 2025) (denoted *PDF* in some papers) compares models via the similarity of the layerwise standard-deviation profiles of their attention parameters. *REEF* (Zhang et al., 2024) computes the centered-kernel-alignment (CKA) similarity between the feature representations elicited from the same samples across two models. *PCS* and *ICS* (Zeng et al., 2024) (referred to collectively as *HuRef* in some papers) constitute weight-similarity methods: PCS flattens all parameters and measures their cosine similarity, whereas ICS constructs invariant terms from the weights and measures cosine similarity upon those invariants. In Section 5.3, we additionally incorporate a (weaker) gradient-based fingerprint, *Gradient* (Shao et al., 2025). Following Zhang et al. (2024), we adopt a **0.8 similarity threshold** for binary decisions.

We note that, throughout all experiment tables, cell colors encode lineage: green designates models sharing a common source, while red designates distinct sources. By way of illustration, green s_{42}^{init} vs. s_{42}^{base} contrasts a model initialized with seed 42 against its continued-pretrained counterpart, and is hence rendered green. Conversely, red s_{2000}^{init} vs. s_{42}^{base} contrasts a seed-2000 initialization with a model trained from a seed-42 initialization, and is hence rendered red. Additionally, ✓ signifies a correct detection, whereas ✗ signifies an error.

5.1 BIRTH-TO-LIFECYCLE “BIOMETRIC” FINGERPRINTING

We train, entirely from scratch, 12-layer, 12-head LLaMA-style models (Touvron et al., 2023) equipped with RoPE (Su et al., 2021) alongside Qwen-style models (Team, 2024). In the main paper we present the results for the LLaMA-style models and relegate those for the Qwen-style models to Appendix D; the overarching conclusions are consistent across the two.

Table 1: Comparison of the fingerprint behaviors exhibited by models initialized with differing seeds.

Model Pairs	p -value (> 0.01)
s_{42}^{init} VS. s_{2000}^{base}	0.116
s_{123}^{init} VS. s_{42}^{base}	1.000
s_{1000}^{init} VS. s_{123}^{base}	0.327
s_{2000}^{init} VS. s_{1000}^{base}	1.000

Table 3: An identical dataset and training order do not render fingerprint behaviors identical across differing initializations.

Model Pair	p -value (> 0.01)
s_{42}^{init} VS. s_{123}^{base}	0.573
s_{123}^{init} VS. s_{2000}^{base}	0.724
s_{2000}^{init} VS. s_{1000}^{base}	0.109
s_{1000}^{init} VS. s_{42}^{base}	0.883

Table 2: Trained models share the very fingerprint behaviors of their initialization (p -value < 0.01).

Model Pairs	SeedPrints p -value	Baselines			
		Intrinsic	REEF	PCS	ICS
s_{42}^{init} VS. s_{42}^{base}	3.37e-26 $\checkmark$	-0.021 $\times$	0.375 $\times$	0.580 $\times$	0.196 $\times$
s_{123}^{init} VS. s_{123}^{base}	3.16e-33 $\checkmark$	0.149 $\times$	0.369 $\times$	0.581 $\times$	0.188 $\times$
s_{1000}^{init} VS. s_{1000}^{base}	1.12e-21 $\checkmark$	-0.252 $\times$	0.381 $\times$	0.581 $\times$	0.188 $\times$
s_{2000}^{init} VS. s_{2000}^{base}	3.70e-31 $\checkmark$	-0.337 $\times$	0.331 $\times$	0.581 $\times$	0.188 $\times$

Table 4: Persistence of the fingerprint under continual training upon diverse datasets (base model: seed 1000, corpus openwebtext).

Setting	SeedPrints p -value	Baselines			
Continual corpus (seed)		Intrinsic	REEF	PCS	ICS
TinyStories (1000)	≈ 0 $\checkmark$	1.000 $\checkmark$	0.759 $\times$	0.999 $\checkmark$	0.996 $\checkmark$
TinyStories (123)	1.000 $\checkmark$	0.950 $\times$	0.658 $\checkmark$	0.332 $\checkmark$	0.012 $\checkmark$
the_stack (1000)	≈ 0 $\checkmark$	0.489 $\times$	0.557 $\times$	0.585 $\times$	0.123 $\times$
the_stack (123)	1.000 $\checkmark$	0.445 $\checkmark$	0.580 $\checkmark$	0.301 $\checkmark$	0.026 $\checkmark$

Different initialization seeds produce distinct fingerprints Table 1 reports the p -values yielded by our correlation tests between pairs of models initialized with different random seeds (42, 123, 1000, and 2000). All p -values are consistently > 0.01 , an outcome indicating that our method reliably discriminates between models trained from different seeds. This demonstrates that distinct seeds engender distinct fingerprint behaviors, thereby permitting models to be separated “at birth.”

Training preserves the initialization fingerprint. Table 2 juxtaposes each initialization model s^{init} with its descendant s^{base} trained on the OpenWebText dataset (Gokaslan et al., 2019) ($\approx 10B$ tokens). Across all seed-model pairs, the p -values are consistently < 0.01 , indicating that their bias profiles remain strongly correlated and hence share a common lineage. In brief, the trained model inherits the very fingerprint of its initialization. We likewise evaluate the baseline methods; without exception, they fail to distinguish across seeds, which in turn intimates that their separability arises from training-induced artifacts rather than from initialization.

Identical data and order do not make fingerprints converge In Table 3, all four “suspicious” models s_i^{base} for $i \in \{42, 123, 100, 2000\}$ are trained on *exactly the same corpus* (OpenWebText) and in an *identical data order* (we fix the training seed so as to lock the data order); the sole point of divergence resides in their initialization seeds i . Across all cross-seed pairs, the p -values remain consistently > 0.01 , in stark contradistinction to the near-zero values reported in Table 2. In other words, fingerprints remain seed-specific even under an identical data distribution and curriculum.

Continual training on diverse datasets does not confound the fingerprint The purpose of our foregoing experiments is solely to exhibit the strengths of SeedPrints, namely its capacity to serve as a biometric fingerprint. From a copyright standpoint, the inability of prior work to distinguish models bearing different seeds does not constitute a weakness, inasmuch as initialization seeds possess no well-defined copyright status. The genuine fragility resides elsewhere: their attribution is readily misled by *data distribution, failing to recognize lineage whenever the training distribution shifts substantially over the course of continued training.*

In Table 4, we continue training a base model (seed 1000, pretrained on OpenWebText (Gokaslan et al., 2019)) upon two markedly disparate datasets: TinyStories (Eldan & Li, 2023) (synthetic children’s stories) and The Stack (Kocetkov et al., 2022) (permissively licensed GitHub code). We contrast (i) true descendants trained from the base against (ii) *distractors* derived from a different base model (seed 123). Since the baseline methods are insensitive to initialization differences, the distractor base is pretrained with a different data order on OpenWebText before being continually trained on the *same* target corpus to constitute the distractor. The question at issue is whether attribution methods can identify which descendant genuinely shares lineage with the base.

Table 5: Fingerprinting results under large-scale finetuning. Each row juxtaposes a target model against LLaMA-2-7B. **SeedPrints** reports the p -value obtained from our correlation test (< 0.01 betokens a strong signal), whereas the four baselines all report similarity scores (threshold = 0.8, with higher values preferable).

Model	# Tokens	SeedPrints p	Intrinsic $\uparrow$	REEF ($\uparrow$)	PCS ($\uparrow$)	ICS ($\uparrow$)
Llama-2-finance-7B (Heenan, 2023)	5M	10^{-41955} ✓	1.0000✓	0.9950✓	0.9979✓	0.9952✓
Vicuna-1.5-7B (Chiang et al., 2023)	370M	$10^{-103043}$ ✓	1.0000✓	0.9985✓	0.9985✓	0.9949✓
Wizardmath-7B (Luo et al., 2023)	1.8B	$10^{-180550}$ ✓	1.0000✓	0.9979✓	1.0000✓	0.9994✓
Meditron-7B (Chen et al., 2023)	48B	10^{-42566} ✓	0.9990✓	0.9978✓	1.0000✓	0.9817✓
CodeLlama-7B (Meta AI, 2024)	500B	10^{-3552} ✓	0.9480✓	0.9947✓	0.6863✗	0.3369✗
Llemma-7B (Azerbayev et al., 2023)	700B	10^{-5136} ✓	0.9470✗	0.9984✓	0.6682✗	0.2905✗

We find that the prior baselines uniformly fail under the code setting (The Stack), misclassifying true descendants as distractors. This betrays the fact that they largely track domain similarity rather than lineage identity: TinyStories lies closer in distribution to the pretraining corpus (OpenWebText), whereas The Stack diverges sharply, and so pronounced a distribution shift can readily elude detection. Our method, in contrast, correctly attributes lineage across both corpora. Our fingerprint is therefore no mere proxy for data distribution: it withstands substantial domain shift and persists well beyond the initial pretraining stage.

5.2 ALL-STAGE VERIFIABLE FINGERPRINTS

Controlled pretraining trajectory. Can our method reliably identify its offspring along the entirety of the training trajectory? We first corroborate this via a controlled pretraining experiment on OpenWebText, in which the initialization seed is known, by testing lineage at intermediate checkpoints; we observe consistent detection across all stages (Appendix D.2.1, Figure 5).

Large-scale pretraining. To appraise lineage under realistic foundation-model training, we further scrutinize ten checkpoints from the OLMo-2-7B (OLMo et al., 2024) Stage-1 run (5B $\rightarrow$ 3.9T tokens), treating the terminal checkpoint as the target model and all antecedent checkpoints as the tested ancestors. Unlike the controlled OpenWebText setting, this stage encompasses heterogeneous corpora, protracted optimization, and large-scale representation drift. We plot $1 - p$ so as to preserve directional consistency with the similarity-based baselines. All p -values are $\ll 0.01$ and numerically proximate to zero.

In Figure 1, we observe that **all existing baselines fail to verify lineage during early pretraining** (e.g., $ICS \approx 0$, $PCS < 0.3$ and $REEF < 0.6$ within the first trillion tokens), whereas SeedPrints remains consistently verifiable and progressively strengthens throughout training. These results throw into relief a critical lacuna in existing evaluations: **most prior work concentrates on verifying lineage only after fine-tuning, precisely the regime in which lineage detection is substantially easier.** Our findings, however, demonstrate that large-scale pretraining itself can dramatically remold representations and fortify lineage signals, thereby engendering a potentially *false sense of safety*. We contend that lineage verification must accord priority to the early stages of pretraining, where misuse is most difficult to detect, rather than resting solely upon late-stage checkpoints.

Large-scale finetuning. We further collate our method against the existing baselines under standard evaluations at the finetuning stage. Specifically, we test suspect models fine-tuned from Llama-2-7b (the base model) with data volumes ranging from 5 million to 700 billion tokens. The suspects encompass a diversity of downstream variants, among them Llama-2-finance-7b (Heenan, 2023), Vicuna-1.5-7b (Chiang et al., 2023), WizardMath-7b (Luo et al., 2023), Chinese-LLaMA-2-7b (Chen et al., 2023), Code-Llama-7b (Meta AI, 2024), and Llemma-7b (Azerbayev et al., 2023), whose respective fine-tuning data volumes amount to 5M, 370M, 1.8B, 13B, 500B, and 700B tokens. As reported in Table 5, our method consistently sustains $p < 0.01$ across the entirety of these settings.

5.3 ROBUSTNESS UNDER REALISTIC DEPLOYMENTS

Real-world deployments routinely subject foundation models to *parameter-altering* adaptations (e.g., fine-tuning, PEFT, quantization) that are liable to attenuate or distort fingerprint signals. Our method is expressly engineered to retain its efficacy within this regime. To assess its reliability under such realistic conditions, we adopt *LeaFBench* (Shao et al., 2025), a benchmark devised for language-model copyright auditing.

Across Source Models. We consider two categories of source model for the purposes of auditing: *Pre-Trained (PT)* and *Instruction-Tuned (IT)*. For each category, the task is to distinguish *derivative* models (procured through post-training adaptations) from *independent* models (trained without any recourse to the source weights). This mirrors the common auditing scenarios in which one must adjudicate lineage claims for either a PT base or its IT variant. Our evaluation subsumes 65 models in total; the complete enumeration of model names and HuggingFace repositories is provided in Appendix E.

Metrics. We appraise detection performance by means of (i) the Area Under the ROC Curve (AUC) and (ii) the Kolmogorov–Smirnov (KS) statistic (Berger & Zhou, 2014), the latter of which gauges the maximal separation between the score distributions of derivative and independent models.

Our method issues per-model p -values obtained from statistical tests, and these differ fundamentally in kind from the similarity scores employed by the baseline methods. Whereas similarity scores are linear measures upon which differences bear a proportional meaning, p -values are tail probabilities that register the rarity of an observation under the null hypothesis and therefore admit no linear-scale interpretation. In order to enable comparison with baseline methods that compute AUC from similarity scores, we convert our p -values into scores via $s = 1 - p$. This conversion is not fully commensurate with the statistical meaning of p -values and may indeed be suboptimal for our method, since small p -values vary on an exponential scale while threshold sweeping proceeds linearly, rendering fine-grained distinctions difficult to resolve. Even so, our approach still attains comparable or superior performance. For evaluation, we (i) compute AUC for all methods by sweeping thresholds over the scores s , and (ii) report the KS statistic as a threshold-free measure of distributional separability. This protocol secures compatibility with the baseline pipelines while conserving the statistical interpretability of our method. We refrain from reporting Manhattan distance, inasmuch as absolute distances between p -values carry no meaningful interpretation.

We note that, in the majority of practical scenarios, the calibration of an optimal similarity threshold is infeasible. While we report AUC for the sake of comparability, our method differs fundamentally from prior approaches in that it furnishes a direct statistical test. **Given any pair of models, our approach emits a p -value that licenses a definitive decision as to whether they share a common lineage, without any reliance upon tunable thresholds**, thereby guaranteeing reliable verification in practice.

Parameter-Altering Techniques. To gauge fingerprinting robustness, we evaluate models produced by way of (1) Instruction tuning (Instruct), (2) General-purpose fine-tuning (Finetune), (3) Parameter-efficient fine-tuning (PEFT), (4) Quantization, (5) Model merging (Merge), and (6) Distillation.

5.3.1 FAMILY-WISE EFFECTIVENESS

There are seven model families in total. For each family, we treat both its pretrained base model (PT) and its instruction-tuned model (IT) as separate source models and compare them against all remaining models in the pool of 65. The results are reported in Table 6. We omit the TinyLlama-1.1B families from Table 6, since they constitute trivial cases for all methods. It bears noting that each column corresponds to a single family—for which AUC is computed over the corresponding tests using a family-calibrated threshold—whereas the overall score is computed across all test pairs with a global threshold; the overall score is thus not equal to the average of the per-family scores. All three methods—*Intrinsic*, *ICS*, and our *SeedPrints*—are essentially saturated across every alteration (with overall AUC of 0.994, 0.994, and 0.992, respectively, and KS of 0.989, 0.943, and 0.986). For these methods, the family-wise scores are, in the preponderance of cases, the full score. The remaining two baselines, *REEF* and *Gradient*, by contrast, perform substantially worse. These results indicate that, although our method is designed primarily to capture intrinsic seed-level fingerprints, it effectively operates as a biometric-like fingerprint affording reliable and persistent identity tracking across a diversity of model families.

5.3.2 ALTERATION-WISE ROBUSTNESS

Across the various alteration types, *Instruct* is consistently the most tractable case (even the weakest baseline, *Gradient*, attains AUC 0.895), whereas *Finetune* and *Merge* prove, on the whole, more formidable for all methods—for *Finetune*, in particular, no method achieves perfect performance. The effects of *PEFT*, *Quantization*, and *Distillation* are more equivocal: *REEF* exhibits notable declines under *Quantization* ($0.996 \rightarrow 0.871$) and *Distillation* ($0.996 \rightarrow 0.858$) while remaining resilient to

Table 6: Comparison of the performance of LLM fingerprinting methods across differing source models. “PT” and “IT” denote, respectively, the use of pre-trained and instruction-tuned models as source models. A per-family breakdown is furnished for the 6 families that possess both PT and IT models. We note that the “Overall” scores are computed over all test model pairs rather than averaged across families.

Method	Metric	Qwen2.5-7B		Qwen2.5-14B		Llama3.1-8B		Mistral-7B		Gemma2-2B		Llama2-7B		Overall
		PT	IT	PT	IT	PT	IT	PT	IT	PT	IT	PT	IT	
REEF	AUC $\uparrow$	0.796	0.862	0.947	0.913	1.000	0.999	0.997	0.997	0.963	0.968	0.750	0.706	0.915
	KS Statistic $\uparrow$	0.492	0.633	0.875	0.735	0.987	0.987	0.975	0.980	0.950	0.837	0.750	0.694	0.739
Gradient	AUC $\uparrow$	0.657	0.743	0.763	0.789	0.897	0.904	0.872	0.682	1.000	1.000	0.650	0.768	0.801
	KS Statistic $\uparrow$	0.354	0.423	0.500	0.563	0.646	0.705	0.725	0.500	1.000	1.000	0.275	0.479	0.508
ICS	AUC $\uparrow$	1.000	1.000	0.997	0.997	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.994
	KS Statistic $\uparrow$	1.000	1.000	0.975	0.979	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.943
Intrinsic	AUC $\uparrow$	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.994
	KS Statistic $\uparrow$	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.989
SeedPrints	AUC $\uparrow$	0.992	0.994	0.988	0.990	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.992
	KS Statistic $\uparrow$	0.985	0.987	0.975	0.980	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.986

Table 7: Performance of LLM fingerprinting methods across a range of parameter-altering techniques.

Method	Metric	Instruct	Finetune	PEFT	Quantization	Merge	Distillation	Overall Avg
REEF	AUC $\uparrow$	0.996	0.872	0.994	0.871	0.954	0.858	0.924
	KS Statistic $\uparrow$	0.949	0.637	0.940	0.812	0.851	0.615	0.801
Gradient	AUC $\uparrow$	0.895	0.729	0.776	0.756	0.832	0.887	0.812
	KS Statistic $\uparrow$	0.756	0.441	0.469	0.586	0.519	0.679	0.575
ICS	AUC $\uparrow$	1.000	0.987	1.000	1.000	0.981	1.000	0.995
	KS Statistic $\uparrow$	1.000	0.925	1.000	1.000	0.941	1.000	0.978
Intrinsic	AUC $\uparrow$	1.000	0.980	1.000	1.000	1.000	1.000	0.997
	KS Statistic $\uparrow$	1.000	0.971	1.000	1.000	1.000	1.000	0.995
SeedPrints	AUC $\uparrow$	1.000	0.995	0.990	1.000	0.959	1.000	0.991
	KS Statistic $\uparrow$	1.000	0.990	0.980	1.000	0.941	1.000	0.985

PEFT (0.994); conversely, *PEFT* and *Quantization* pose greater difficulty for Gradient (0.895 $\rightarrow$ 0.776/0.756). The three stronger methods remain, for the most part, robust to all transformations (with every score > 0.9). SeedPrints declines marginally to 0.959 on *Merge*, faltering on a solitary merge model out of seventeen. Although this amounts to but a single misclassification, it intimates that weight-space interpolation with unrelated model weights may potentially dilute the initialization signal below the detection threshold. Averaged across all six alteration types, Intrinsic still achieves the highest overall AUC, followed closely by ICS and SeedPrints.

5.4 ABLATION STUDY ON HYPERPARAMETERS

Our method is governed by three hyperparameters: the number of random input sequences n , the length of each random sequence l , and the fingerprint size m . The first two jointly control the overall volume of random inputs, whereas m stipulates how many low-mean coordinates are enlisted for testing. Throughout the main experiments of Sections 5.2 and 5.3, we set $n=2000$, $l=1024$, and $m=0.1d_{\text{out}}$, where d_{out} denotes the hidden-state dimensionality, thereby rendering the method adaptable to different open-source models. Here we vary each hyperparameter in turn while holding the remainder fixed at their default values (namely $n = 2000$, $m = 400$, $l = 768$), and we report the empirical accuracy and false-positive rates under a fixed significance level $\alpha = 0.01$ in Table 8.

We observe that increasing n and l consistently ameliorates identification performance. This tendency is elucidated by our theoretical analysis in Section C.2, which establishes that, in order to recover the underlying population-level bias accurately, the requisite number of queries must satisfy $n = \Omega\left(\frac{\log d_{\text{out}}}{\gamma^2}\right)$, where γ denotes the bias margin separating the biased coordinates from the remaining dimensions. It follows that a target output dimension demands a minimal number of samples for accurate detection, which accounts for the suboptimal performance incurred under smaller n . On the other hand, γ grows with l owing to accumulation effects, so that longer sequences reinforce the separation. Consequently, under a fixed (and insufficient) number of sequences, increasing l diminishes the required query count and thereby improves detection.

The effect of m , by contrast, is non-monotonic. When m is too small, the selected coordinates are unstable, for want of sufficient empirical observations to constitute a reliable distribution. When m is too large, the inclusion of numerous irrelevant coordinates introduces noise that contracts γ , which would in turn demand quadratically more samples by way of compensation. Empirically, a choice of m within the range 200–400 performs well across model sizes and comports with our heuristic of $m \approx 0.1d_{\text{out}}$ for typical $d_{\text{out}} \in [3000, 5000]$.

6 CONCLUSION

In this work, we introduced *SeedPrints*, a stronger and more intrinsic notion of LLM fingerprinting that traces a model’s lineage all the way back to its random initialization. We demonstrated that initialization itself impresses a persistent “Galtonian” fingerprint upon neural language models: untrained models already exhibit stable, seed-dependent token-preference patterns, and these biases endure—statistically and measurably—throughout training. Building upon this observation, we proposed a distribution-correlation test over *identity dimensions* that detects shared lineage with calibrated significance. Across LLaMA- and Qwen-style families and a broad suite of realistic deployments (encompassing continued pretraining on divergent corpora, instruction tuning, PEFT, quantization, merging, and distillation), SeedPrints enables “birth-to-lifecycle” verification and remains robust under domain shift, thereby furnishing a practical instrument for provenance and copyright auditing.

Limitations and Future Directions This work sets out to uncover fingerprint signals rooted in initialization-stage bias. The proposed method presupposes access to model outputs (e.g., logits or hidden states) and may not be applicable when only restricted API access is afforded. We stress, nonetheless, that the pursuit of *true fingerprints* for LLMs remains at an incipient stage: most existing approaches—and black-box methods in particular—rely upon signals that emerge only in the later training stages or from alignment-driven artifacts, and are therefore incapable of robustly conferring protection at the pretrained stage. Reliable all-stage fingerprints are, in consequence, all the more urgently required. See the full discussions in Section A.2.

Table 8: Ablation study under significance level $\alpha = 0.01$.

Ablation on n	200	500	2000
ACC	0.7368	0.7778	0.9375
Empirical FPR	0	0	0
Ablation on l	512	768	1024
ACC	0.9048	0.9375	1.0000
Empirical FPR	0.04	0	0
Ablation on m	200	400	800
ACC	0.9375	0.9375	0.875
Empirical FPR	0.04	0	0.1463

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A LLM USAGE AND LIMITATION DISCUSSIONS

A.1 LLM USAGE

We employ AI assistants—namely ChatGPT and Gemini—for writing and formatting support. Their use extends to grammar and style checking, the refinement of figure and table captions for clarity, and other surface-level edits. For programming-related tasks, we occasionally enlist GitHub Copilot and Claude as coding assistants, for instance in code auto-completion and the provision of debugging hints.

A.2 LIMITATIONS AND FUTURE DIRECTIONS

Although the present work uncovers seed-born, training-persistent biases that can serve as fingerprint signals, our method remains circumscribed within the white-box paradigm. It may prove inapplicable when only restricted API access is afforded—for instance, when responses are rendered deterministic or are safety-filtered. We stress, nevertheless, that the pursuit of *true fingerprints* for LLMs remains at a decidedly early stage, even within the white-box setting. Most existing fingerprinting approaches rely upon training-induced signatures or alignment behaviors, and are thus incapable of robustly protecting pretrained or early-stage models. By analogy to human fingerprints, which remain stable irrespective of age or environment, we hold that a true LLM fingerprint ought to persist across continued training and deployment contexts rather than to reflect incidental training artifacts. Achieving this—particularly under black-box access, where a substantial performance gap relative to white-box methods yet persists—constitutes a challenging open direction that will require considerable future work. We hope that this work galvanizes further research toward the establishment of principled, robust fingerprinting frameworks for large models.

B VALIDATION OF THE EMPIRICAL AND ANALYTICAL NULL DISTRIBUTION

Our hypothesis test in Section 4 adjudicates whether two models share lineage by comparing the distribution of their correlation statistics over random inputs against a null distribution. The null distribution models the correlation statistics obtaining between two *independent models evaluated on random inputs*.

We consider two approaches to approximating this null distribution: (1) an empirical simulation-based null, obtained by applying the full pipeline to Gaussian surrogate outputs, and (ii) an analytical null, derived from the known distribution of Kendall–Tau correlations. In this section, we vindicate these approximations by way of three steps: (i) characterizing the true null using independently initialized models, (ii) introducing the Gaussian surrogate as a tractable simulation-based proxy, and (iii) validating the analytical approximation.

B.1 EMPIRICAL TRUE NULL FROM INDEPENDENTLY INITIALIZED MODELS

The ideal null distribution corresponds to the correlation statistics between two independent models evaluated on random inputs. The most pristine means of instantiating this independence is to compare two independently initialized models (bearing different random seeds and having undergone no training). We estimate this empirical null by evaluating ~ 2500 such pairs and computing Kendall–Tau correlations on 10,000 random inputs (as in Section 3). Figure 3 demonstrates that the true null distribution is, empirically, a Gaussian centered about zero.

B.2 GAUSSIAN SURROGATE AS A SIMULATION-BASED APPROXIMATION

The direct estimation of the null distribution using independently initialized models is computationally expensive, inasmuch as it presupposes access to a great many models together with repeated evaluations. To secure a tractable alternative, we introduce a simulation-based null in which Gaussian surrogates stand as proxies for model outputs under the null.

This choice is motivated by two considerations. First, the approximation of neural-network outputs by Gaussian distributions is a widely adopted expedient in the auditing literature, buttressed by both empirical and theoretical studies (Carlini et al., 2022; Lee et al., 2019). Second, our pipeline hinges

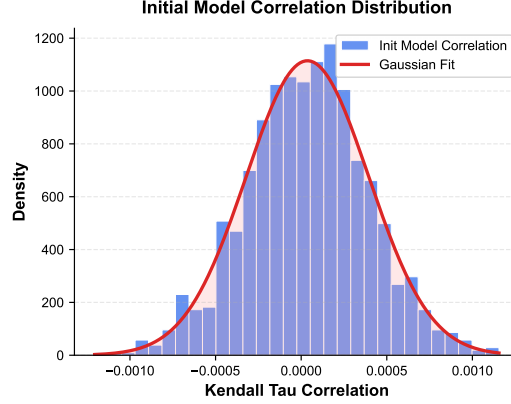


Figure 3: The empirical correlation distribution over ~ 2500 pairs of independently initialized models, evaluated on random inputs, closely conforms to a Gaussian distribution centered at zero.

only upon relative ordering and selection operations (e.g., rank correlations and bottom- m selection) and does not draw upon any semantic structure in the inputs. Moreover, randomly initialized models possess independent parameter matrices, which suggests that their outputs under the null exhibit no strong structured dependencies. Sampling independent Gaussian matrices therefore furnishes a simple and tractable proxy for modeling the null behavior of independently initialized models.

We construct the surrogate null by sampling Gaussian matrices and applying the full pipeline, encompassing both identity-dimension selection and correlation computation:

1. Sample two Gaussian matrices of shape $\mathbb{R}^{N \times d_{\text{out}}}$,
2. Extract the most biased dimensions by selecting the bottom- m ranked average dimensions,
3. Take the intersection set $\mathcal{S}$ and compute correlations on it.

This simulation-based construction applies the identical selection procedure to Gaussian samples and thereby captures, as a matter of course, the dependency patterns introduced by the pipeline itself (e.g., those arising from the selection of $\mathcal{S}$ across dimensions). As displayed in Figure 4 (middle), the resulting distribution coincides with the empirical null obtained from independently initialized models with respect to its centered Gaussian shape, which indicates that the Gaussian surrogate constitutes an effective simulation-based approximation of the true null.

B.3 ANALYTICAL APPROXIMATION UNDER WEAK DEPENDENCE

While the Gaussian surrogate captures the behavior of the null distribution under the full pipeline, it nonetheless necessitates repeated sampling and injects randomness into the estimation process. We therefore additionally propose the most efficient of the alternatives, an analytical approximation.

Under the null hypothesis that the two models are independent², each Kendall–Tau correlation τ_j is approximately zero-mean with known variance σ^2 . Were the correlations across identity dimensions independent, the central limit theorem would imply

$$\bar{\tau} = \frac{1}{|\mathcal{S}|} \sum_{j \in \mathcal{S}} \tau_j \sim \mathcal{N}\left(0, \frac{\sigma^2}{|\mathcal{S}|}\right).$$

In practice, softmax normalization introduces dependencies across dimensions. Nevertheless, Figure 4 demonstrates that the analytical Gaussian closely coincides with the distribution obtained from the Gaussian surrogate pipeline across a range of settings, including varying numbers of random inputs n and output dimensions D . This suggests that such dependencies are sufficiently weak as not to affect materially the accuracy of the approximation. A plausible explanation is that the use of a

²As discussed in Section B.2, although the outputs may exhibit weak dependencies on account of shared inputs, this approximation remains accurate empirically.

comparatively high temperature ($T = 10$) smooths the output distribution and attenuates the coupling between dimensions.

Taken together, these results vindicate the use of the analytical Gaussian null in Section 4 as an efficient, deterministic, and accurate approximation, one that obviates the need for costly empirical baseline construction.

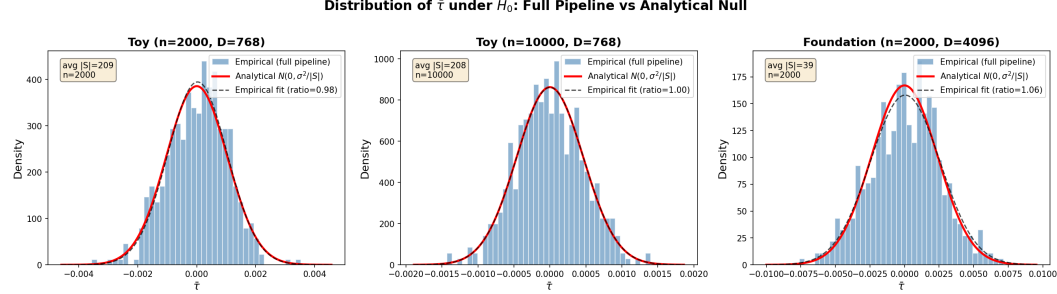


Figure 4: Validation of the analytical Gaussian null. We apply the full pipeline (encompassing identity-dimension selection and correlation aggregation) to Gaussian surrogate outputs and compare the resulting distribution of $\bar{t}$ against the analytical Gaussian $\mathcal{N}(0, \sigma^2/|S|)$. The close correspondence indicates that the analytical approximation accurately captures the null behavior, the weak dependencies introduced by the pipeline notwithstanding.

On variance scaling. The true null obtained from randomly initialized models possesses a slightly smaller variance than the Gaussian null approximation. This discrepancy, however, does not perturb our hypothesis-test results. Our primary evaluation for the simulation-based null in Section D.2 relies upon the Mann–Whitney U-test, which, being rank-based, is invariant to any positive scaling: for any $c > 0$, $x_i < x_j \Leftrightarrow cx_i < cx_j$. All rank statistics and rejection thresholds thus remain unaltered. For those tests that do depend upon the absolute variance, we have additionally evaluated variance-sensitive statistics such as the t -test, whose statistic scales as $t' = t/c$ as the variance decreases. Since $c < 1$ in our case, the scaling diminishes the magnitude of t and thereby renders the test more conservative rather than inflating significance. Empirically, across the extensive experiments of Section D.2, we find that the variance scaling does not impair the power of the t -test, which indicates that the distributional separation between the alternative hypothesis and the null substantially exceeds the scaled variance.

For the analytical null adopted in the main paper, the variance is incorporated explicitly into the normalization of the z -score, so that any scale discrepancy is absorbed within the denominator. Since the analytical variance is slightly larger than its empirical counterpart, the resulting z -scores are smaller, which renders the test conservative and forestalls false positives.

C THEORETICAL PROPERTIES OF OUR METHODS

Our objective is to identify the output dimensions that the initialized model is least disposed to predict under random inputs. These dimensions correspond to directions that the model inherently disfavors by virtue of initialization-induced bias (independent of the input distribution), and they constitute the basis of a stable fingerprint signal.

Setup. Consider a model g with fixed parameters together with random inputs x_i (uniform random tokens or random embeddings, as in the main paper). For each output dimension $j \in [d_{\text{out}}]$, $g_j(x)$ is accordingly a random variable whose population mean is $\bar{g}_j := \mathbb{E}[g_j(x)]$. The population-level directions we seek to capture correspond to the m output dimensions of smallest expected value, formally defined as

$$\mathcal{M} = \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \bar{g}_j.$$

Let $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{\text{out}})})$ denote the sorted means in non-decreasing order, and write

$$\Gamma(m) := \bar{g}_{(m+1)} - \bar{g}_{(m)} > 0$$

for the population “gap” between the m -th and $(m+1)$ -th smallest means. This gap quantifies how well separated the target set $\mathcal{M}$ is from the remaining coordinates. Crucially, for a fixed m and input distribution, $\Gamma(m)$ depends solely upon the model’s inherent initialization bias. *Increasing the length of the random input sequences, however, amplifies this bias via concentration, thereby rendering the separation more pronounced at the population level.*

There are two natural empirical strategies for extracting such population-level least-likely dimensions:

1. **Sample-mean-based extraction (ours).** We estimate the expectation of each output coordinate by the empirical mean over random inputs $X = \{x_i\}_{i=1}^n$ and thereupon select the m coordinates of smallest value. Following the notation of Section 4, in which g denotes the model (either the base model f or the suspect model f'):

$$\hat{g} = \frac{1}{n} \sum_{i=1}^n g(x_i), \quad \widehat{\mathcal{M}}^{\text{mean}} = \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \hat{g}_j.$$

2. **Per-sample argmin voting.** For each random input $x_i \in \mathbb{R}^{\ell \times d}$, we select the smallest-value coordinate and thereafter retain the m most frequent dimensions:

$$j^*(x_i) = \arg \min_{j \in \{1, \dots, d_{\text{out}}\}} g_j(x_i), \quad \widehat{\mathcal{M}}^{\text{vote}} = \arg \max_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{i=1}^n \mathbf{1}\{j^*(x_i) \in J\}.$$

Let $p_j := \Pr(j^*(x) = j)$ denote the population probability that index j attains the minimum, and let

$$\mathcal{M}_{\text{vote}} := \arg \max_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} p_j$$

be the population top- m set under voting.

Both approaches are intuitively reasonable. As we establish below, however, the sample-mean-based strategy is provably stable, whereas per-sample argmin voting is discontinuous and liable to flip under arbitrarily small perturbations, which renders it inherently noise-sensitive.

C.1 STABILITY ANALYSIS OF MEAN-BASED IDENTITY EXTRACTION VS. PER-SAMPLE ARGMIN VOTING

C.1.1 STABILITY OF SAMPLE-MEAN-BASED EXTRACTION

We first establish that the sample-mean-based operator, which selects the m smallest coordinates of the empirical mean vector, is stable under small perturbations to the model outputs. Such perturbations represent realistic variations in the outputs of one and the *same* model—for instance, numerical noise or evaluation under differing precision modes.

Formally, we model the perturbed outputs as $g'(x) = g(x) + \eta(x)$, where $\eta(x) \in \mathbb{R}^{d_{\text{out}}}$ represents the stochastic deviations. We assume that for each sample x_i and coordinate j , the random variables $\eta_j(x_i)$ are independent across samples (not necessarily across coordinates) and σ -sub-Gaussian with $\mathbb{E}[\eta_j(x_i)] = 0$.³

Lemma C.1 (Stability of sample-mean-based extraction). *Let $\hat{g}$ and $\hat{g}'$ be the empirical means over $X = \{x_i\}_{i=1}^n$, i.e., $\hat{g} = \frac{1}{n} \sum_{i=1}^n g(x_i)$ and $\hat{g}' = \frac{1}{n} \sum_{i=1}^n g'(x_i) = \hat{g} + \frac{1}{n} \sum_{i=1}^n \eta(x_i)$. Let $\widehat{\Gamma}(m)$ denote the empirical gap at rank m . Then,*

$$\Pr(\widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g)) \leq 2d_{\text{out}} \exp\left(-\frac{n\widehat{\Gamma}(m)^2}{8\sigma^2}\right).$$

Interpretation. The lemma furnishes a quantitative upper bound on the probability that the sample-mean-based extractor alters its selected set under perturbation. This instability probability decays exponentially in the number of averaged samples n , owing to the variance reduction from σ^2 per sample to σ^2/n in the empirical mean.

³This sub-Gaussian assumption is a mild one: bounded random variables are σ -sub-Gaussian (by Hoeffding’s lemma), and common initialization schemes, in concert with bounded activations, yield light-tailed behavior in both initialized and trained networks. The Gaussian case $\eta_j(x_i) \sim \mathcal{N}(0, \sigma^2)$ is subsumed as a special instance of this assumption.

Proof. We condition upon the sample $X = \{x_i\}_{i=1}^n$ throughout the proof, so that $\hat{g}$ and $\hat{\Gamma}(m)$ are held fixed and all probabilities are taken over the randomness of $\{\eta(x_i)\}_{i=1}^n$ alone.

For any coordinate j , by the definition of $\hat{g}'$, we have $\hat{g}'_j - \hat{g}_j = \frac{1}{n} \sum_{i=1}^n \eta_j(x_i)$. Each $\eta_j(x_i)$ is mean-zero and σ -sub-Gaussian, whence their average is mean-zero and $(\sigma/\sqrt{n})$ -sub-Gaussian: the variance proxy scales as $1/n$ under averaging. Consequently, for any $\varepsilon > 0$, the standard sub-Gaussian tail bound yields

$$\Pr(|\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X) \leq 2 \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right). \quad (1)$$

Now consider the event that *some* coordinate fluctuates by more than ε :

$$\left\{ \max_{1 \leq j \leq d_{\text{out}}} |\hat{g}'_j - \hat{g}_j| \geq \varepsilon \right\}.$$

A union bound taken over all d_{out} coordinates yields

$$\Pr\left(\max_j |\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X\right) \leq \sum_{j=1}^{d_{\text{out}}} \Pr(|\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X) \leq 2d_{\text{out}} \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right). \quad (2)$$

We now show that

$$\left\{ \widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g) \right\} \subseteq \left\{ \max_j |\hat{g}'_j - \hat{g}_j| \geq \hat{\Gamma}(m)/2 \right\}.$$

Equivalently, whenever $\max_j |\hat{g}'_j - \hat{g}_j| < \hat{\Gamma}(m)/2$, the bottom- m identity set remains unchanged.

Let $j_{(1)}, \dots, j_{(d_{\text{out}})}$ be a permutation of $\{1, \dots, d_{\text{out}}\}$ for which $\hat{g}_{j_{(1)}} \leq \hat{g}_{j_{(2)}} \leq \dots \leq \hat{g}_{j_{(d_{\text{out}})}}$. Recall that the empirical gap at rank m is $\hat{\Gamma}(m) := \hat{g}_{j_{(m+1)}} - \hat{g}_{j_{(m)}} > 0$. Denote $\widehat{\mathcal{M}}^{\text{mean}}(g) := \{j_{(1)}, \dots, j_{(m)}\}$ and $\widehat{\mathcal{M}}_c^{\text{mean}}(g) := \{j_{(m+1)}, \dots, j_{(d_{\text{out}})}\}$. For any $i \in \widehat{\mathcal{M}}^{\text{mean}}(g)$ and any $\ell \in \widehat{\mathcal{M}}_c^{\text{mean}}(g)$, we have $\hat{g}_\ell - \hat{g}_i \geq \hat{g}_{j_{(m+1)}} - \hat{g}_{j_{(m)}} = \hat{\Gamma}(m)$. Given that $|\hat{g}'_i - \hat{g}_i| < \hat{\Gamma}(m)/2$ and $|\hat{g}'_\ell - \hat{g}_\ell| < \hat{\Gamma}(m)/2$, it follows that

$$\hat{g}'_\ell - \hat{g}'_i \geq (\hat{g}_\ell - \frac{\hat{\Gamma}(m)}{2}) - (\hat{g}_i + \frac{\hat{\Gamma}(m)}{2}) \geq \hat{\Gamma}(m) - \hat{\Gamma}(m) = 0.$$

Thus, under this deviation bound, no coordinate in $\widehat{\mathcal{M}}_c^{\text{mean}}(g)$ can become smaller than any coordinate in $\widehat{\mathcal{M}}^{\text{mean}}(g)$, and therefore

$$\widehat{\mathcal{M}}^{\text{mean}}(g') = \widehat{\mathcal{M}}^{\text{mean}}(g).$$

Since $\hat{\Gamma}(m)$ is held fixed under conditioning on X , substituting $\varepsilon = \hat{\Gamma}(m)/2$ into (2) yields

$$\Pr\left(\widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g)\right) \leq \Pr\left(\max_j |\hat{g}'_j - \hat{g}_j| \geq \hat{\Gamma}(m)/2 \mid X\right) \leq 2d_{\text{out}} \exp\left(-\frac{n\hat{\Gamma}(m)^2}{8\sigma^2}\right).$$

□

C.1.2 INSTABILITY OF PER-SAMPLE ARGMIN VOTING

Per-sample voting, by contrast, applies a discontinuous winner-take-all operator to each input. Arbitrarily small perturbations of the outputs may therefore alter decisions with non-vanishing probability. We continue to consider the same stochastic perturbation model $g'(x) = g(x) + \eta(x)$.

Lemma C.2 (Instability of per-sample argmin voting). *For a fixed input x , let*

$$j^*(x; g) := \arg \min_j g_j(x), \quad \Delta_g(x) := \min_{j \neq j^*(x; g)} \{g_j(x) - g_{j^*(x; g)}(x)\} > 0$$

denote the argmin and its margin. Then, for any fixed x ,

$$\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right),$$

and for n i.i.d. inputs $x_1, \dots, x_n$,

$$\Pr\left(\exists i \leq n : j^*(x_i; g') \neq j^*(x_i; g)\right) \geq 1 - (1 - \alpha)^n,$$

where $\alpha := \mathbb{E}_x \left[\Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right) \right]$.

Interpretation. In contradistinction to mean extraction, voting derives no benefit from variance reduction: each input entails a direct, hard decision predicated on noisy outputs. Increasing n merely multiplies the opportunities for index-flips.

Proof. We first bound the per-input change probability $\Pr(j^*(x; g') \neq j^*(x; g) \mid x)$. Fix x and write $j^* := j^*(x; g)$ for the sake of brevity. By the definition of the margin, $\Delta_g(x) = \min_{j \neq j^*} \{g_j(x) - g_{j^*}(x)\}$. Let $j^{(2)}$ denote an index attaining this minimum, i.e., $g_{j^{(2)}}(x) - g_{j^*}(x) = \Delta_g(x)$.

Under the perturbation $g'(x) = g(x) + \eta(x)$, we have

$$g'_{j^{(2)}}(x) - g'_{j^*}(x) = [g_{j^{(2)}}(x) - g_{j^*}(x)] + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] = \Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)].$$

Since $\eta_{j^{(2)}}(x)$ and $\eta_{j^*}(x)$ are independent $\mathcal{N}(0, \sigma^2)$ variables, their difference is Gaussian with

$$\eta_{j^{(2)}}(x) - \eta_{j^*}(x) \sim \mathcal{N}(0, 2\sigma^2).$$

The event that the argmin changes, $j^*(x; g') \neq j^*(x; g)$, assuredly occurs whenever $g'_{j^{(2)}}(x) \leq g'_{j^*}(x)$. Hence,

$$\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Pr(g'_{j^{(2)}}(x) \leq g'_{j^*}(x) \mid x).$$

Using the expression above,

$$\Pr(g'_{j^{(2)}}(x) \leq g'_{j^*}(x) \mid x) = \Pr(g'_{j^{(2)}}(x) - g'_{j^*}(x) \leq 0 \mid x) = \Pr(\Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] \leq 0 \mid x).$$

Let $Z \sim \mathcal{N}(0, 1)$. Since $\eta_{j^{(2)}}(x) - \eta_{j^*}(x) \stackrel{d}{=} \sqrt{2}\sigma Z$, we obtain

$$\Pr(\Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] \leq 0 \mid x) = \Pr\left(Z \leq -\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right) = \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right).$$

Therefore, $\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right)$.

Now consider n i.i.d. inputs $x_1, \dots, x_n$. Conditional on the inputs, the events

$$E_i := \{j^*(x_i; g') \neq j^*(x_i; g)\} \quad E_i^c := \{j^*(x_i; g') = j^*(x_i; g)\}$$

are independent across i , inasmuch as the perturbations $\eta(x_i)$ are independent across i . Moreover,

$$\Pr(E_i \mid x_i) = \Pr(j^*(x_i; g') \neq j^*(x_i; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x_i)}{\sqrt{2}\sigma}\right) \geq \Phi\left(-\frac{\Delta_g(x_i)}{\sqrt{2}\sigma}\right).$$

Define $q(x_i) := \Pr(j^*(x_i; g') \neq j^*(x_i; g) \mid x)$, and $\alpha := \mathbb{E}_x\left[\Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right)\right]$. Taking expectations over the randomness of x_i , we obtain $\mathbb{E}[\Pr(E_i \mid x_i)] = \alpha$. Therefore, by independence across i ,

$$\Pr(E_1^c \cap \dots \cap E_n^c) = \mathbb{E}\left[\prod_{i=1}^n (1 - q(x_i))\right] \leq \mathbb{E}\left[\prod_{i=1}^n (1 - \Phi(-\Delta_g(x_i)/(\sqrt{2}\sigma)))\right] \leq (1 - \alpha)^n,$$

wherein the last inequality invokes the i.i.d. inputs together with Jensen's inequality applied to the convex function $u \mapsto \log(1 - u)$ on $u \in [0, 1)$. Therefore,

$$\Pr(\exists i \leq n : j^*(x_i; g') \neq j^*(x_i; g)) = 1 - \Pr(E_1^c \cap \dots \cap E_n^c) \geq 1 - (1 - \alpha)^n.$$

Finally, we note that the empirical voting set $\widehat{\mathcal{M}}^{\text{vote}}$ is determined by the empirical counts of the dimensions $\{j^*(x_i; g)\}_{i=1}^n$. Should the empirical gap between the m -th and $(m+1)$ -th most frequent dimensions be of order $O(1/n)$ (i.e., a mere constant number of votes), then a change in $j^*(x_i)$ for a single input x_i suffices to transpose the ordering across this boundary and thereby to alter the top- m set. Combining this observation with the lower bound $1 - (1 - \alpha)^n$ establishes that, in such regimes, the probability that $\widehat{\mathcal{M}}^{\text{vote}}(g') \neq \widehat{\mathcal{M}}^{\text{vote}}(g)$ is bounded away from zero uniformly in n ; that is, per-sample argmin voting reaps no variance reduction from additional samples. $\square$

C.2 INFLUENCE OF HYPERPARAMETERS

We now analyze how many random inputs are requisite for the empirical mean estimator to recover the true bias. We reuse the notation of **Setup**: each output coordinate $g_j(x)$ is a random variable under random inputs $x \sim \mathcal{D}$, with population mean $\bar{g}_j := \mathbb{E}[g_j(x)]$. $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{\text{out}})})$ denotes the sorted means in non-decreasing order, and $\gamma := \bar{g}_{(m+1)} - \bar{g}_{(m)} > 0$ is the separation margin. We estimate $\bar{g}$ from n i.i.d. random inputs:

$$\hat{g}_j := \frac{1}{n} \sum_{i=1}^n g_j(x_i), \quad \hat{g} = (\hat{g}_1, \dots, \hat{g}_{d_{\text{out}}}).$$

The target dimension set is $\mathcal{M}$, of cardinality m , and our empirical estimation yields $\widehat{\mathcal{M}}$.

Lemma C.3 (Sufficient condition for recovering the true set). *If $\|\hat{g} - \bar{g}\|_\infty \leq \varepsilon$ and $2\varepsilon < \gamma$, then the empirical bottom- m set equals the true set: $\widehat{\mathcal{M}} = \mathcal{M}$.*

Proof. For any $j \in \mathcal{M}$ and $j' \notin \mathcal{M}$, we have $\bar{g}_j \leq \bar{g}_{(m)}$ and $\bar{g}_{j'} \geq \bar{g}_{(m+1)}$. The ℓ_∞ bound gives

$$\hat{g}_j \leq \bar{g}_{(m)} + \varepsilon, \quad \hat{g}_{j'} \geq \bar{g}_{(m+1)} - \varepsilon.$$

Thus,

$$\hat{g}_j - \hat{g}_{j'} \leq -(\gamma - 2\varepsilon) < 0,$$

so no outside index can enter nor inside index leave the bottom- m set. $\square$

We next construe this recovery condition in terms of the hyperparameters. We again adopt the same sub-Gaussian assumption as in Section C.1. This assumption is a mild one: by Hoeffding's inequality, any bounded random variable is σ -sub-Gaussian with σ proportional to its range. In our setting, each $g_j(x)$ corresponds to a logit or a final hidden activation, quantities that are typically bounded or light-tailed in both initialized networks (by virtue of controlled parameter initialization) and trained networks.

Corollary C.4 (Sample complexity under sub-Gaussian coordinates). *Suppose each $g_j(x)$ is σ -sub-Gaussian. Then with probability at least $1 - \delta$,*

$$\|\hat{g} - \bar{g}\|_\infty \leq \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

Consequently, if

$$n \geq \frac{8\sigma^2}{\gamma^2} \log\left(\frac{2d_{\text{out}}}{\delta}\right),$$

then $\widehat{\mathcal{M}} = \mathcal{M}$ with probability at least $1 - \delta$.

Proof. For fixed j , sub-Gaussian concentration implies

$$\Pr(|\hat{g}_j - \bar{g}_j| > \varepsilon) \leq 2 \exp(-n\varepsilon^2/2\sigma^2).$$

A union bound over all d_{out} coordinates yields

$$\Pr\left(\|\hat{g} - \bar{g}\|_\infty > \varepsilon\right) \leq \sum_{j=1}^{d_{\text{out}}} \Pr(|\hat{g}_j - \bar{g}_j| > \varepsilon) \leq 2d_{\text{out}} \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right).$$

Set the right-hand side to δ and solve for ε to obtain, with probability at least $1 - \delta$,

$$\|\hat{g} - \bar{g}\|_\infty \leq \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

Now choose

$$\varepsilon := \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

If $n \geq \frac{8\sigma^2}{\gamma^2} \log\left(\frac{2d_{\text{out}}}{\delta}\right)$ (i.e., if $2\varepsilon < \gamma$), then Lemma C.3 applies and implies $\widehat{\mathcal{M}} = \mathcal{M}$ with probability at least $1 - \delta$. $\square$

Table 9: Fingerprint behavior under differing random initializations across model families.

(a) LLaMA-style models					(b) Qwen-style models				
Seed Pair	Logits Output		Hidden State		Seed Pair	Logits Output		Hidden State	
	<i>t-test</i>	<i>U-test</i>	<i>t-test</i>	<i>U-test</i>		<i>t-test</i>	<i>U-test</i>	<i>t-test</i>	<i>U-test</i>
s_{42} vs. s_{2000}	0.404	0.456	0.357	0.532	s_{123} vs. s_{1000}	0.741	0.727	0.094	0.074
s_{123} vs. s_{42}	0.214	0.295	0.678	0.565	s_{1000} vs. s_{123}	0.954	0.971	0.125	0.094
s_{1000} vs. s_{123}	0.219	0.246	0.363	0.335	s_{42} vs. s_{2000}	0.273	0.360	0.451	0.529
s_{2000} vs. s_{1000}	0.282	0.291	0.434	0.481	s_{2000} vs. s_{42}	0.215	0.206	0.230	0.295

Table 10: (LLaMA-style models) Trained models share the very fingerprint behaviors of their initialization (p -value < 0.01).

Model Pair	Logits Output		Hidden State		Baselines			
	<i>t-test</i>	<i>u-test</i>	<i>t-test</i>	<i>u-test</i>	Intrinsic	REEF	PCS	ICS
s_{42}^{init} vs. s_{42}^{base}	3.33e-3✓	1.02e-3✓	2.20e-8✓	6.28e-8✓	-0.021×	0.375×	0.580×	0.196×
s_{123}^{init} vs. s_{123}^{base}	2.06e-3✓	7.33e-3✓	7.09e-6✓	1.37e-5✓	0.149×	0.369×	0.581×	0.188×
s_{1000}^{init} vs. s_{1000}^{base}	2.44e-3✓	4.14e-3✓	5.58e-4✓	2.81e-3✓	-0.252×	0.381×	0.581×	0.188×
s_{2000}^{init} vs. s_{2000}^{base}	5.63e-3✓	6.76e-3✓	4.00e-10✓	1.27e-9✓	-0.337×	0.331×	0.581×	0.188×

Accurate recovery therefore requires $n = \Omega\left(\frac{\log d_{out}}{\gamma^2}\right)$, which is to say that the number of random queries grows only logarithmically with the output dimension while scaling inversely with the squared separation margin γ . The margin γ is itself governed by two contributing factors:

- **Intrinsic bias strength.** Longer random input sequences amplify the initialization-induced bias via concentration, thereby widening the population separation between coordinates.
- **Choice of m .** Distinct choices of m correspond to distinct portions of the ranked spectrum $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{out})})$ and hence yield distinct effective margins in practice.

The former is amenable to control through the input length l , whereas the latter is selected empirically on the basis of the observed gaps within the mean spectrum.

D MORE EXPERIMENT DETAILS

D.1 IMPLEMENTATION DETAILS AND LICENSES

We train all models by means of the Hugging Face Transformers Trainer (Wolf et al., 2020), employing Accelerate (Gugger et al., 2022) for the distributed runs. All open-source models are loaded from their official Hugging Face releases and used in accordance with their original licenses: the Llama models under the Meta Llama Community License, and the remaining models under Apache-2.0. All datasets are downloaded via the Hugging Face Datasets library (the library itself being Apache-2.0), and the dataset content abides by each dataset’s stated license.

D.2 COMPLEMENTARY SEEDPRINTS RESULTS USING SIMULATION-BASED NULL

This section furnishes the complementary results for Section 5. We report results obtained with the empirical simulation-based null of Algorithm 1 under both *logits* and *hidden state* outputs, and we conduct the testing by means of both the one-sided t -test (t -test) and the Mann–Whitney U test (U test).

D.2.1 BIRTH-TO-LIFECYCLE “BIOMETRIC” FINGERPRINTING

We train, entirely from scratch, 12-layer, 12-head LLaMA-style models (Touvron et al., 2023) equipped with RoPE (Su et al., 2021) alongside Qwen-style models (Team, 2024). Because the simulation-based random baseline is stochastic, we report p -values averaged over 10 independent

Table 11: (Qwen-style models) Trained models share the very fingerprint behaviors of their initialization models (p -value < 0.01).

Model Pair	Logits Output		Hidden State	
	t -test	U -test	t -test	U -test
s_{123}^{init} vs. s_{123}^{base}	2.38e-03	1.68e-03	7.36e-15	3.38e-13
s_{1000}^{init} vs. s_{1000}^{base}	1.35e-04	4.84e-05	4.41e-13	2.01e-11
s_{42}^{init} vs. s_{42}^{base}	1.39e-03	1.46e-03	1.06e-24	2.05e-19
s_{2000}^{init} vs. s_{2000}^{base}	1.80e-03	1.28e-03	4.87e-24	1.92e-20

Table 12: Persistence of the fingerprint under continual training upon diverse datasets (base model: seed 1000, corpus openwebtext). U -test designates the Mann–Whitney U test.

(a) LLaMA-style models

Setting	Ours (logits)		Ours (hidden)		Baselines			
Continual corpus (seed)	t -test	U -test	t -test	U -test	Intrinsic	REEF	PCS	ICS
TinyStories (1000)	0 [✓]	0 [✓]	0 [✓]	7.77e-89 [✓]	1.000 [✓]	0.759 [×]	0.999 [✓]	0.996 [✓]
TinyStories (123)	1.000 [✓]	1.00 [✓]	0.943 [✓]	0.902 [✓]	0.950 [×]	0.658 [✓]	0.332 [✓]	0.012 [✓]
the_stack (1000)	0 [✓]	1.73e-287 [✓]	0 [✓]	3.09e-69 [✓]	0.489 [×]	0.557 [×]	0.585 [×]	0.123 [×]
the_stack (123)	0.616 [✓]	0.479 [✓]	0.732 [✓]	0.831 [✓]	0.445 [✓]	0.580 [✓]	0.301 [✓]	0.026 [✓]

(b) Qwen-style models

Setting	Ours (logits)		Ours (hidden)		Baselines			
Continual corpus (seed)	t -test	U -test	t -test	U -test	Intrinsic	REEF	PCS	ICS
TinyStories (1000)	5.36e-09 [✓]	1.92e-07 [✓]	8.49e-214 [✓]	5.09e-71 [✓]	1.000 [✓]	0.957 [✓]	0.999 [✓]	0.996 [✓]
TinyStories (123)	0.434 [✓]	0.433 [✓]	0.256 [✓]	0.065 [✓]	0.913 [×]	0.199 [✓]	0.328 [✓]	0.039 [✓]
the_stack (1000)	0 [✓]	4.16e-237 [✓]	1.16e-211 [✓]	2.30e-76 [✓]	0.999 [✓]	0.313 [×]	0.995 [✓]	0.976 [✓]
the_stack (123)	0.999 [✓]	0.993 [✓]	0.610 [✓]	0.491 [✓]	0.916 [×]	0.255 [✓]	0.328 [✓]	0.038 [✓]

trials and adopt a significance level of $\alpha = 0.01$. It bears emphasis that the absolute magnitude of exceedingly small p -values carries no meaningful interpretation: once p subsides beneath numerical and sampling noise (e.g., $< 10^{-20}$), values such as 10^{-260} ought not be construed as stronger evidence than 10^{-20} , since both decisively reject the null.

Different initialization seeds produce distinct fingerprints Table 9a and Table 9b report the p -values yielded by our correlation tests between pairs of models initialized with different random seeds (42, 123, 1000, and 2000) under the t -test and U -test, for the Llama-style and Qwen-style models respectively. All p -values are consistently > 0.01 , an outcome indicating that our method reliably discriminates between models trained from different seeds. This demonstrates that distinct seeds engender distinct fingerprint behaviors, thereby permitting models to be separated “at birth.”

Training preserves the initialization fingerprint. Table 10 (LLaMA-style) and Table 11 (Qwen-style) juxtapose each initialization model s^{init} with its descendant s^{base} trained on the OpenWebText dataset (Gokaslan et al., 2019) (≈ 10 B tokens). Across all seed–model pairs, the p -values are consistently < 0.01 , indicating that their bias profiles remain strongly correlated and hence share a common lineage. In brief, the trained model inherits the very fingerprint of its initialization; training does not efface the initialization fingerprint. We likewise evaluate the baseline methods in Table 10; without exception, they fail to distinguish across seeds, which in turn intimates that their separability arises from training-induced artifacts rather than from initialization.

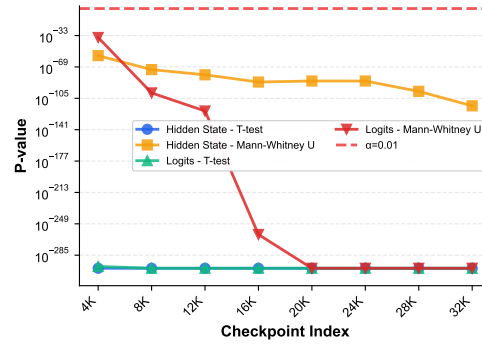
Identical data and order do not make fingerprints converge In Table 13a (LLaMA-style models) and Table 13b (Qwen-style models), all four “suspicious” models s_i^{base} for $i \in \{42, 123, 100, 2000\}$ are trained on *exactly the same corpus (OpenWebText) and in an identical data order* (we fix the training seed so as to lock the data order); the sole point of divergence resides in their initialization

Table 13: An identical dataset and training order do not render fingerprint behaviors identical across differing initializations.

(a) LLaMA-style models					(b) Qwen-style models				
Model Pair	Logits Output		Hidden State		Model Pair	Logits Output		Hidden State	
	<i>t</i> -test	<i>U</i> -test	<i>t</i> -test	<i>U</i> -test		<i>t</i> -test	<i>U</i> -test	<i>t</i> -test	<i>U</i> -test
s_{123}^{init} vs. s_{1000}^{base}	0.484	0.500	0.385	0.486	s_{123}^{init} vs. s_{1000}^{base}	0.598	0.638	0.286	0.254
s_{1000}^{init} vs. s_{2000}^{base}	0.946	0.956	0.135	0.096	s_{1000}^{init} vs. s_{123}^{base}	0.804	0.805	0.236	0.240
s_{42}^{init} vs. s_{123}^{base}	0.598	0.589	0.426	0.337	s_{42}^{init} vs. s_{2000}^{base}	0.589	0.608	0.226	0.2043
s_{2000}^{init} vs. s_{42}^{base}	0.756	0.781	0.388	0.287	s_{2000}^{init} vs. s_{42}^{base}	0.523	0.482	0.312	0.323



(a) LLaMA-style models



(b) Qwen-style models

Figure 5: The fingerprint verifies lineage at every checkpoint (p -values < 0.01) across model families.

seeds i . Our aim is to ascertain whether fingerprint behavior might be effaced or confounded by identical data and order. Across all cross-seed pairs, the p -values remain consistently > 0.01 , in stark contradistinction to the near-zero values of Table 10. That is, fingerprints remain seed-specific even under an identical data distribution and curriculum.

Continual training on diverse datasets does not confound the fingerprint The purpose of our foregoing experiments is solely to exhibit the strengths of SeedPrints, namely its capacity to serve as a biometric fingerprint. From a copyright standpoint, the inability of prior work to distinguish models bearing different seeds does not constitute a weakness, inasmuch as initialization seeds possess no well-defined copyright status. The genuine fragility resides elsewhere: their attribution is readily misled by *data distribution, failing to recognize lineage whenever the training distribution shifts substantially over the course of continued training.*

In Table 12a and Table 12b, we continue training a base model (seed 1000, pretrained on OpenWebText (Gokaslan et al., 2019)) upon two markedly disparate datasets: TinyStories (Eldan & Li, 2023) (synthetic children’s stories) and The Stack (Kocetkov et al., 2022) (permissively licensed GitHub code). We contrast (i) true descendants trained from the base against (ii) *distractors* derived from a different base model (initialized with seed 123 and trained with a different data order on OpenWebText), which is thereafter continued on the *same* corpus. The question at issue is whether attribution methods can identify which descendant genuinely shares lineage with the base.

We find that the prior baselines uniformly fail under the code setting (The Stack), misclassifying true descendants as distractors. This betrays the fact that they largely track domain similarity rather than lineage identity: TinyStories lies closer in distribution to the pretraining corpus (OpenWebText), whereas The Stack diverges sharply, and so pronounced a distribution shift can readily elude detection. Our method, in contrast, correctly attributes lineage across both corpora. Our fingerprint is therefore no mere proxy for data distribution: it withstands substantial domain shift and persists well beyond the initial pretraining stage.

Table 14: Cross-size similarity between Qwen-2.5-14B (the target) and the remaining Qwen-2.5 models.

	Qwen-2.5-0.5B	Qwen-2.5-1.5B	Qwen-2.5-3B	Qwen-2.5-7B
SeedPrint ($\uparrow$)	0.7184	0.9707	0.3898	0.5429
REEF ($\downarrow$)	0.8250	0.8399	0.8833	0.8461
HUREF ($\downarrow$)	-6.37×10^{-5}	-2.84×10^{-5}	-8.70×10^{-6}	-6.08×10^{-6}

D.2.2 ALL-STAGE VERIFIABLE FINGERPRINTS

Controlled pretraining trajectory. We first conduct a controlled pretraining experiment on OpenWebText, in which the initialization seed is known to us. Each intermediate checkpoint is treated as the base, and we test whether our method can reliably identify its offspring along the same training trajectory. All variants consistently recognize the suspect model as belonging to the same lineage, with the p -values remaining beneath the 0.01 threshold (Figure 5a for LLaMA-style models and Figure 5b for Qwen-style models). This corroborates the stability of SeedPrints under continuous optimization.

D.3 CROSS-SIZE FINGERPRINT WITHIN THE QWEN-2.5 SERIES

To examine further whether SeedPrints “collapses” within a model family as the parameter size is varied, we report the cross-size similarity between Qwen-2.5-14B (the target) and smaller models drawn from the same Qwen-2.5 family. Table 14 presents the results for SeedPrint ($\uparrow$) alongside two baselines, REEF ($\downarrow$) and HUREF ($\downarrow$).

To probe whether initialization-born fingerprints “collapse” into a family-level signature as model size is scaled, we conduct a cross-size study within the Qwen-2.5 series. We treat Qwen-2.5-14B as the protected target model and consider Qwen-2.5- $\{0.5B, 1.5B, 3B, 7B\}$ as suspect models that, while belonging to the same architectural family, are not trained as descendants of Qwen-2.5-14B. As discussed in the main paper, SeedPrints emits a p -value for the hypothesis test “the suspect shares initialization-born fingerprints with the target,” whereas REEF and HUREF are similarity-based baselines that directly output continuous scores and thus oblige the user to select a threshold.

Several observations merit remark. First, all SeedPrints p -values sit comfortably above a typical significance level (e.g., $\alpha=0.01$). Under our hypothesis-testing view, this signifies that none of the smaller Qwen-2.5 models is flagged as sharing a lineage with Qwen-2.5-14B. Such behavior is precisely what is desired in our setting: even though these models belong to the same family and reuse a similar architectural design, the fingerprinting test does not collapse into a single family-level signature that would spuriously conflate distinct pre-training runs.

The cross-size p -values occupy disparate regions of the $[0, 1]$ interval rather than concentrating about a single value. We do *not* construe these magnitudes as a continuous “similarity score”—their function is to underwrite a calibrated yes/no decision at a fixed significance level. The observed spread nonetheless indicates that our test statistic remains sensitive to the concrete conjunction of architecture and initialization seed, rather than degenerating into a generic pattern common to all Qwen-2.5 variants.

Finally, the similarity-based baselines comport themselves differently. REEF produces scores confined to a relatively narrow band (0.8250–0.8833) across all sizes, whereas HUREF outputs values numerically very close to zero (on the order of 10^{-5}). Since these methods provide no built-in decision rule, operators must select thresholds by hand, and it remains unclear how one might robustly separate cross-size variants from genuinely lineage-related models by means of a single cutoff. SeedPrints, in contrast, directly yields a statistical decision at a prescribed significance level, thereby affording a clearer and more operational answer to the question “do these two models plausibly share a training lineage?” within a model family.

E MODEL CATALOG AND DECODING SETTINGS

Our evaluation encompasses **58** models in total. In what follows we document the decoding configuration employed throughout all runs and enumerate the model catalog grouped by family. For each derived model we further indicate the post-training transformation (*finetune*, *adapter*, *merge*, *quantization*, *distillation*). “PT” and “IT” denote, respectively, *pre-trained* and *instruction-tuned* source models.

Parameter	Value
max_new_tokens	512
temperature	0.7
top_p	0.9
top_k	50
do_sample	true
max_input_length	512

E.1 MODEL LIST BY FAMILY

E.1.1 QWEN-2.5-7B

Variant	HuggingFace repo	Type
PT	Qwen/Qwen2.5-7B	—
IT	Qwen/Qwen2.5-7B-Instruct	—
Derived	Qwen/Qwen2.5-Math-7B	finetune
Derived	Qwen/Qwen2.5-Coder-7B-Instruct	finetune
Derived	WangCa/Qwen2.5-7B-Medicine	finetune
Derived	huihui-ai/Qwen2.5-7B-Instruct-abliterated-v2	finetune
Derived	Locutusque/StockQwen-2.5-7B	merge
Derived	bunnycore/QevaCoT-7B-Stock	merge
Derived	fangcaotank/task-10-Qwen-Qwen2.5-7B-Instruct	adapter
Derived	SeeFlock/task-12-Qwen-Qwen2.5-7B-Instruct	adapter
Derived	Qwen/Qwen2.5-7B-Instruct-GPTQ-Int4	quantization
Derived	Qwen/Qwen2.5-7B-Instruct-GPTQ-Int8	quantization
Derived	Lansechen/Qwen2.5-7B-Open-R1-Distill	distillation

E.1.2 QWEN2.5-14B

Variant	HuggingFace repo	Type
PT	Qwen/Qwen2.5-14B	—
IT	Qwen/Qwen2.5-14B-Instruct	—
Derived	Qwen/Qwen2.5-Coder-14B	finetune
Derived	oxyapi/oxy-1-small	finetune
Derived	v000000/Qwen2.5-14B-Gutenberg-Instruct-Slerpeno	merge
Derived	ToastyPigeon/qwen-story-test-qlora	adapter
Derived	Qwen/Qwen2.5-14B-Instruct-GPTQ-Int4	quantization
Derived	deepseek-ai/DeepSeek-R1-Distill-Qwen-14B	distillation

E.1.3 LLAMA-3.1-8B

Variant	HuggingFace repo	Type
PT	meta-llama/Llama-3.1-8B	—
IT	meta-llama/Llama-3.1-8B-Instruct	—
Derived	ValiantLabs/Llama3.1-8B-Fireplace2	finetune
Derived	RedHatAI/Llama-3.1-8B-tldr	finetune
Derived	proxectonos/Llama-3.1-Carballo	finetune
Derived	mlabonne/Meta-Llama-3.1-8B-Instruct-abliterated	finetune
Derived	gaverfraxz/Meta-Llama-3.1-8B-Instruct-HalfAbliterated-TIES	merge
Derived	Xiaojian9992024/Llama3.1-8B-ExtraMix	merge
Derived	LlamaFactoryAI/Llama-3.1-8B-Instruct-cv-job-description-matching	adapter
Derived	chchen/Llama-3.1-8B-Instruct-PsyCourse-fold7	adapter
Derived	iqbalamo93/Meta-Llama-3.1-8B-Instruct-GPTQ-Q_8	quantization
Derived	DaraV/LLaMA-3.1-8B-Instruct-INT4-GPTQ	quantization
Derived	asas-ai/Llama-3.1-8B-Instruct-Open-R1-Distill	distillation

E.1.4 MISTRAL-7B-v0.3

Variant	HuggingFace repo	Type
PT	mistralai/Mistral-7B-v0.3	—
IT	mistralai/Mistral-7B-Instruct-v0.3	—
Derived	KurmaAI/AQUA-7B	finetune
Derived	openfoodfacts/spellcheck-mistral-7b	finetune
Derived	grimjim/Mistral-7B-Instruct-demi-merge-v0.3-7B	merge
Derived	chaymaemrhrioui/mistral-Brain_Model_ACC_Trainer	adapter
Derived	RedHatAI/Mistral-7B-Instruct-v0.3-GPTQ-4bit	quantization
Derived	eganwo/mistral7b-distilled-from-deepseek-r1-qwen32b	distillation

E.1.5 GEMMA-2-2B

Variant	HuggingFace repo	Type
PT	google/gemma-2-2b	—
IT	google/gemma-2-2b-it	—
Derived	rinna/gemma-2-baku-2b	finetune
Derived	anakin87/gemma-2-2b-neogenesis-ita	finetune
Derived	vonjack/gemma2-2b-merged	merge
Derived	google-cloud-partnership/gemma-2-2b-it-lora-sql	adapter
Derived	qilowoq/gemma-2-2B-it-4Bit-GPTQ	quantization
Derived	Syed-Hasan-8503/Gemma-2-2b-it-distilled	distillation

E.1.6 LLAMA-2-7B

Variant	HuggingFace repo	Type
PT	meta-llama/Llama-2-7b-hf	—
IT	meta-llama/Llama-2-7b-chat-hf	—
Derived	allenai/tulu-2-7b	finetune
Derived	QIAIUNCC/EYE-Llama_qa	finetune
Derived	DevQuasar/coma-7B-v0.1	merge
Derived	Ammar-1/llama2-Better-Tune	adapter
Derived	TheBloke/Llama-2-7B-Chat-GPTQ	quantization
Derived	cygu/llama-2-7b-logit-watermark-distill-kgw-k1-gamma0.25-delta2	distillation

E.1.7 TINYLLAMA-1.1B

Variant	HuggingFace repo	Type
PT/IT	TinyLlama/TinyLlama-1.1B-Chat-v1.0	—
Derived	alexredna/TinyLlama-1.1B-Chat-v1.0-reasoning-v2	finetune
Derived	Edentns/DataVortexTL-1.1B-v0.1	finetune
Derived	appvoid/dot-v2.7	merge
Derived	barissg1c/tinyllama-tarot-v1	adapter
Derived	TheBloke/TinyLlama-1.1B-Chat-v1.0-GPTQ	quantization
Derived	anudaw/distilled-code-llama	distillation

Notes. “Type” designates the post-training transformation relative to the PT/IT source model: *finetune* subsumes supervised and preference-optimization variants; *adapter* subsumes LoRA/QLoRA-style modules; *merge* subsumes model soups and TIES-style merges; *quantization* subsumes GPTQ/INTx variants; and *distillation* subsumes student models distilled from reasoning-augmented teachers.

E.2 SAMPLE RESULTS

We present partial results from a single LeafBench evaluation run in Table 15 (*u*-test with the simulation-based null). The reported metrics are $1 - p$, employed as similarity scores. We adopt a significance level of 0.01 for the *p*-values; empirically, LeafBench identifies an optimal decision threshold of 0.9920, which lends further support to the reliability of our method.

Crucially, although individual *p*-values ought not be construed as linear similarity measures, their distribution across model pairs is nonetheless informative: same-lineage models yield markedly lower *p*-values, whereas unrelated models remain approximately uniform.

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Model	Qwen-2.5-7B	Qwen-2.5-7B-Instruct	Qwen2.5-14B	Qwen2.5-14B-Instruct
Qwen-2.5-7B	1.000	1.000	0.079	0.000
Qwen-2.5-7B-Instruct	1.000	1.000	0.813	0.536
Qwen2.5-7B-Math	1.000	0.994	0.047	0.987
Qwen2.5-7B-Coder	0.996	0.999	0.177	0.002
Qwen2.5-7B-Instruct-Medicine	1.000	1.000	0.576	0.369
Qwen2.5-7B-Instruct-Abiliterated	1.000	1.000	0.276	0.886
Qwen2.5-7B-Stock	1.000	1.000	0.939	0.225
QevaCoT-7B	1.000	1.000	0.199	0.119
Qwen2.5-7B-Instruct-Task-10	1.000	1.000	0.672	0.090
Qwen2.5-7B-Instruct-Task-12	1.000	1.000	0.155	0.987
Qwen2.5-7B-Instruct-Int4	1.000	1.000	0.827	0.961
Qwen2.5-7B-Instruct-Int8	1.000	1.000	0.914	0.739
Qwen2.5-7B-Open-R1-Distill	1.000	1.000	0.302	0.955
Qwen2.5-14B	0.324	0.487	1.000	1.000
Qwen2.5-14B-Instruct	0.403	0.702	1.000	1.000
Qwen2.5-Coder-14B	0.492	0.660	1.000	1.000
oxy-1-small	0.492	0.104	1.000	1.000
Qwen2.5-14B-Gutenberg-Instruct-Slerpeno	0.856	0.931	1.000	1.000
Qwen-story-test-qlora	0.112	0.898	1.000	1.000
Qwen2.5-14B-Instruct-GPTQ-Int4	0.583	0.092	1.000	1.000
DeepSeek-R1-Distill-Qwen-14B	0.204	0.000	1.000	1.000
Llama-3.1-8B	0.638	0.714	0.156	0.000
Llama-3.1-8B-Instruct	0.643	0.118	0.128	0.510
Llama-3.1-8B-Fireplace2	0.715	0.532	0.065	0.354
Llama-3.1-8B-TLDR	0.925	0.811	0.331	0.014
Llama-3.1-8B-Carballo	0.001	0.605	0.003	0.946
Llama-3.1-8B-Instruct-Abliterated	0.061	0.005	0.121	0.422
Llama-3.1-8B-Instruct-HalfAbliterated-TIES	0.356	0.509	0.796	0.000
Llama-3.1-8B-ExtraMix	0.640	0.160	0.721	0.875
Llama-3.1-8B-Instruct-cv-job-description-matching	0.684	0.292	0.004	0.001
Llama-3.1-8B-Instruct-PsyCourse-fold7	0.682	0.057	0.063	0.191
Llama-3.1-8B-Instruct-8bit	0.798	0.214	0.154	0.385
Llama-3.1-8B-Instruct-4bit	0.513	0.023	0.005	0.933
Llama-3.1-8B-Instruct-Open-R1-Distill	0.800	0.101	0.772	0.024
Mistral-7B-v0.3	0.224	0.352	0.506	0.044
Mistral-7B-v0.3-Instruct	0.000	0.000	0.984	0.104
AQUA-7B	0.014	0.613	0.703	0.569
Mistral-7B-v0.3-Spellcheck	0.005	0.481	0.872	0.105
Mistral-7B-v0.3-Instruct-demi-merge	0.904	0.774	0.058	0.522
Mistral-7B-v0.3-Brain	0.340	0.983	0.481	0.016
Mistral-7B-v0.3-Instruct-GPTQ-4bit	0.002	0.281	0.089	0.784
Mistral-7B-distilled-from-deepseek-r1-qwen32b	0.009	0.001	0.511	0.015
Gemma-2-2b	0.747	0.869	0.689	0.000
Gemma-2-2b-it	0.944	0.039	0.626	0.060
Gemma-2-baku-2b	0.616	0.231	0.176	0.464
Gemma-2-2b-neogenesis-ita	0.002	0.536	0.079	0.980
Gemma-2-2b-merged	0.951	0.012	0.651	0.075
Gemma-2-2b-it-lora-sql	0.236	0.408	0.000	0.000
Gemma-2-2B-it-4Bit-GPTQ	0.945	0.159	0.000	0.008
Gemma-2-2b-it-distilled	0.482	0.573	0.271	0.276
Llama-2-7b	0.443	0.470	0.148	0.276
Llama-2-7b-chat	0.528	0.868	0.424	0.766
tulu-2-7b	0.125	0.272	0.170	0.888
EYE-Llama_qa	0.657	0.244	0.674	0.968
coma-7B-v0.1	0.953	0.004	0.573	0.731
llama2-Better-Tune	0.136	0.699	0.469	0.860
Llama-2-7B-Chat-GPTQ	0.594	0.603	0.582	0.424
llama-2-7b-logit-watermark-distill-kgw-k1-gamma0.25-delta2	0.349	0.224	0.366	0.976

Table 15: **Sample results.** Pairwise similarity between the target models (rows) and the base models (columns). Cells for which $1 - p \geq 0.99$ are highlighted in green to denote same-lineage pairs.